

# PL/SQL & SQL Coding Guidelines



**Tips for Development & Operation**

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# About

## Foreword



In the I.T. world of today, robust and secure applications are becoming more and more important. Many business processes no longer work without I.T. and the dependence of businesses on their I.T. has grown tremendously, meaning we need robust and maintainable applications. An important requirement is to have standards and guidelines, which make it possible to maintain source code created by a number of people quickly and easily. This forms the basis of well functioning off- or on-shoring strategy, as it allows quality assurance to be carried out efficiently at the source.

Good standards and guidelines are based on the wealth of experience and knowledge gained from past (and future?) problems, such as those, which can arise in a cloud environment, for example.

A handwritten signature in black ink, reading "h. Lankes".

Urban Lankes  
President of the board of directors  
Trivadis



The Oracle Database Developer community is made stronger by resources freely shared by experts around the world, such as the Trivadis Coding Guidelines. If you have not yet adopted standards for writing SQL and PL/SQL in your applications, this is a great place to start.

A handwritten signature in black ink, reading "Steven Feuerstein".

Steven Feuerstein  
Team Lead, Oracle Developer Advocates  
Oracle



Coding Guidelines are a crucial part of software development. It is a matter of fact, that code is more often read than written – therefore we should take efforts to ease the work of the reader, which is not necessarily the author.

I am convinced that this standard may be a good starting point for your own guidelines.

Roger Troller  
Senior Consultant  
finnova AG Bankware

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## Revision History

The first version of these guidelines was compiled by Roger Troller on March 17, 2009. Jörn Kulessa, Daniela Reiner, Richard Bushnell, Andreas Flubacher and Thomas Mauch helped Roger complete version 1.2 until August 21, 2009. This was the first GA version. The handy printed version in A5 format was distributed free of charge at the DOAG Annual Conference and on other occasions. Since then Roger updated the guidelines regularly. Philipp Salvisberg was involved in the review process for version 3.0 which was a major update. Philipp took the lead, after Roger left Trivadis in 2016.

Since July, 7 2018 these guidelines are hosted on GitHub. Ready to be enhanced by the community and forked to fit specific needs.

On <https://github.com/Trivadis/plsql-and-sql-coding-guidelines/releases> you find the release information for every version since 1.2.

# Introduction

This document describes rules and recommendations for developing applications using the PL/SQL & SQL Language.

## Scope

This document applies to the PL/SQL and SQL language as used within ORACLE databases and tools, which access ORACLE databases.

## Document Conventions

SQALE (Software Quality Assessment based on Lifecycle Expectations) is a method to support the evaluation of a software application source code. It is a generic method, independent of the language and source code analysis tools.

### SQALE characteristics and subcharacteristics

| Characteristic  | Description and Subcharacteristics                                                                                                                                                                                                                                                                                                                                             |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Changeability   | <p>The capability of the software product to enable a specified modification to be implemented.</p> <ul style="list-style-type: none"><li>• Architecture related changeability</li><li>• Logic related changeability</li><li>• Data related changeability</li></ul>                                                                                                            |
| Efficiency      | <p>The capability of the software product to provide appropriate performance, relative to the amount of resources used, under stated conditions.</p> <ul style="list-style-type: none"><li>• Memory use</li><li>• Processor use</li><li>• Network use</li></ul>                                                                                                                |
| Maintainability | <p>The capability of the software product to be modified. Modifications may include corrections, improvements or adaptation of the software to changes in environment, and in requirements and functional specifications.</p> <ul style="list-style-type: none"><li>• Understandability</li><li>• Readability</li></ul>                                                        |
| Portability     | <p>The capability of the software product to be transferred from one environment to another.</p> <ul style="list-style-type: none"><li>• Compiler related portability</li><li>• Hardware related portability</li><li>• Language related portability</li><li>• OS related portability</li><li>• Software related portability</li><li>• Time zone related portability.</li></ul> |
| Reliability     | <p>The capability of the software product to maintain a specified level of performance when used under specified conditions.</p> <ul style="list-style-type: none"><li>• Architecture related reliability</li><li>• Data related reliability</li></ul>                                                                                                                         |



|             |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <ul style="list-style-type: none"> <li>• Exception handling</li> <li>• Fault tolerance</li> <li>• Instruction related reliability</li> <li>• Logic related reliability</li> <li>• Resource related reliability</li> <li>• Synchronization related reliability</li> <li>• Unit tests coverage.</li> </ul>                                                                                                                     |
| Reusability | <p>The capability of the software product to be reused within the development process.</p> <ul style="list-style-type: none"> <li>• Modularity</li> <li>• Transportability.</li> </ul>                                                                                                                                                                                                                                       |
| Security    | <p>The capability of the software product to protect information and data so that unauthorized persons or systems cannot read or modify them and authorized persons or systems are not denied access to them.</p> <ul style="list-style-type: none"> <li>• API abuse</li> <li>• Errors (e.g. leaving a system in a vulnerable state)</li> <li>• Input validation and representation</li> <li>• Security features.</li> </ul> |
| Testability | <p>The capability of the software product to enable modified software to be validated.</p> <ul style="list-style-type: none"> <li>• Integration level testability</li> <li>• Unit level testability.</li> </ul>                                                                                                                                                                                                              |

## Severity of the rule

|                                                                                                     |                                                              |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <b>Blocker</b>  | Will or may result in a bug.                                 |
|  <b>Critical</b> | Will have a high/direct impact on the maintenance cost.      |
|  <b>Major</b>    | Will have a medium/potential impact on the maintenance cost. |
|  <b>Minor</b>    | Will have a low impact on the maintenance cost.              |
|  <b>Info</b>     | Very low impact; it is just a remediation cost report.       |

## Keywords used

| Keyword     | Meaning                                                                         |
|-------------|---------------------------------------------------------------------------------|
| Always      | Emphasizes this rule must be enforced.                                          |
| Never       | Emphasizes this action must not happen.                                         |
| Avoid       | Emphasizes that the action should be prevented, but some exceptions may exist.  |
| Try         | Emphasizes that the rule should be attempted whenever possible and appropriate. |
| Example     | Precedes text used to illustrate a rule or a recommendation.                    |
| Reason      | Explains the thoughts and purpose behind a rule or a recommendation.            |
| Restriction | Describes the circumstances to be fulfilled to make use of a rule.              |

## Why are standards important

For a machine executing a program, code formatting is of no importance. However, for the human eye, well-formatted code is much easier to read. Modern tools can help to implement format and coding rules.

Implementing formatting and coding standards has the following advantages for PL/SQL development:

- Well-formatted code is easier to read, analyze and maintain (not only for the author but also for other developers).
- The developers do not have to define their own guidelines - it is already defined.
- The code has a structure that makes it easier to avoid making errors.
- The code is more efficient concerning performance and organization of the whole application.
- The code is more modular and thus easier to use for other applications.

## We have other standards

This document only defines possible standards. These standards are not written in stone, but are meant as guidelines. If standards already exist, and they are different from those in this document, it makes no sense to change them.

## We do not agree with all your standards

There are basically two types of standards.

### 1. Non-controversial

These standards make sense. There is no reason not to follow them. An example of this category is [G-2150](#): Avoid comparisons with NULL value, consider using IS [NOT] NULL.

### 2. Controversial

Almost every rule/guideline falls into this category. An example of this category is [3 space indentation](#). - Why not 2 or 4 or even 8? Why not use tabs? You can argue in favor of all these options. In most cases it does not really matter which option you choose. Being consistent is more important. In this case it will make the code easier to read.

For very controversial rules, we have started to include the reasoning either as a footnote or directly in the text.

Usually it is not helpful to open an issue on GitHub to request to change a highly controversial rule such as the one mentioned. For example, use 2 spaces instead of 3 spaces for an indentation. This leads to a discussion where the people in favor of 4 spaces start to argue as well. There is no right or wrong here. You just have to agree on a standard.

More effective is to fork [this repository](#) and amend the standards to fit your needs/expectations.

# Naming Conventions

## General Guidelines

1. Never use names with a leading numeric character.
2. Always choose meaningful and specific names.
3. Avoid using abbreviations unless the full name is excessively long.
4. Avoid long abbreviations. Abbreviations should be shorter than 5 characters.
5. Any abbreviations must be widely known and accepted.
6. Create a glossary with all accepted abbreviations.
7. Never use ORACLE keywords as names. A list of ORACLEs keywords may be found in the dictionary view `V$RESERVED_WORDS`.
8. Avoid adding redundant or meaningless prefixes and suffixes to identifiers.  
Example: `CREATE TABLE emp_table`.
9. Always use one spoken language (e.g. English, German, French) for all objects in your application.
10. Always use the same names for elements with the same meaning.

## Naming Conventions for PL/SQL

In general, ORACLE is not case sensitive with names. A variable named `personname` is equal to one named `PersonName`, as well as to one named `PERSONNAME`. Some products (e.g. TMDA by Trivadis, APEX, OWB) put each name within double quotes (") so ORACLE will treat these names to be case sensitive. Using case sensitive variable names force developers to use double quotes for each reference to the variable. Our recommendation is to write all names in lowercase and to avoid double quoted identifiers.

A widely used convention is to follow a `{prefix}variablecontent{suffix}` pattern.

The following table shows a possible set of naming conventions.

| Identifier                   | Prefix | Suffix | Example           |
|------------------------------|--------|--------|-------------------|
| Global Variable              | g_     |        | g_version         |
| Local Variable               | l_     |        | l_version         |
| Cursor                       | c_     |        | c_employees       |
| Record                       | r_     |        | r_employee        |
| Array / Table                | t_     |        | t_employees       |
| Object                       | o_     |        | o_employee        |
| Cursor Parameter             | p_     |        | p_empno           |
| In Parameter                 | in_    |        | in_empno          |
| Out Parameter                | out_   |        | out_ename         |
| In/Out Parameter             | io_    |        | io_employee       |
| Record Type Definitions      | r_     | _type  | r_employee_type   |
| Array/Table Type Definitions | t_     | _type  | t_employees_type  |
| Exception                    | e_     |        | e_employee_exists |
| Constants                    | co_    |        | co_empno          |
| Subtypes                     |        | _type  | big_string_type   |

## Database Object Naming Conventions

Never enclose object names (table names, column names, etc.) in double quotes to enforce mixed case or lower case object names in the data dictionary.

### Collection Type

A collection type should include the name of the collected objects in their name. Furthermore, they should have the suffix `_ct` to identify it as a collection.

Optionally prefixed by a project abbreviation.

Examples:

- `employees_ct`
- `orders_ct`

## Column

Singular name of what is stored in the column (unless the column data type is a collection, in this case you use plural <sup>1</sup> names)

Add a comment to the database dictionary for every column.

## Check Constraint

Table name or table abbreviation followed by the column and/or role of the check constraint, a `_ck` and an optional number suffix.

Examples:

- `employees_salary_min_ck`
- `orders_mode_ck`

## DML / Instead of Trigger

Choose a naming convention that includes:

either

- the name of the object the trigger is added to,
- any of the triggering events:
  - `_br_iud` for Before Row on Insert, Update and Delete
  - `_io_id` for Instead of Insert and Delete

or

- the name of the object the trigger is added to,
- the activity done by the trigger,
- the suffix `_trg`

Examples:

- `employees_br_iud`
- `orders_audit_trg`
- `orders_journal_trg`

## Foreign Key Constraint

Table abbreviation followed by referenced table abbreviation followed by a `_fk` and an optional number suffix.

Examples:

- `empl_dept_fk`
- `sct_icmd_ic_fk1`

## Function

Name is built from a verb followed by a noun in general. Nevertheless, it is not sensible to call a function `get_...` as a function always gets something.

The name of the function should answer the question "What is the outcome of the function?"

Optionally prefixed by a project abbreviation.

Example: `employee_by_id`

If more than one function provides the same outcome, you have to be more specific with the name.

## Index

Indexes serving a constraint (primary, unique or foreign key) are named accordingly.

Other indexes should have the name of the table and columns (or their purpose) in their name and should also have `_idx` as a suffix.

## Object Type

The name of an object type is built by its content (singular) followed by a `_ot` suffix.

Optionally prefixed by a project abbreviation.

Example: `employee_ot`

## Package

Name is built from the content that is contained within the package.

Optionally prefixed by a project abbreviation.

Examples:

- `employees_api` - API for the employee table
- `logging_up` - Utilities including logging support

## Primary Key Constraint

Table name or table abbreviation followed by the suffix `_pk`.

Examples:

- `employees_pk`
- `departments_pk`
- `sct_contracts_pk`

## Procedure

Name is built from a verb followed by a noun. The name of the procedure should answer the question “What is done?”

Procedures and functions are often named with underscores between words because some editors write all letters in uppercase in the object tree, so it is difficult to read them.

Optionally prefixed by a project abbreviation.

Examples:

- `calculate_salary`
- `set_hiredate`
- `check_order_state`

## Sequence

Name is built from the table name (or its abbreviation) the sequence serves as primary key generator and the suffix `_seq` or the purpose of the sequence followed by a `_seq`.

Optionally prefixed by a project abbreviation.

Examples:

- `employees_seq`
- `order_number_seq`

## Synonym

Synonyms should be used to address an object in a foreign schema rather than to rename an object. Therefore, synonyms should share the name with the referenced object.

## System Trigger

Name of the event the trigger is based on.

- Activity done by the trigger
- Suffix `_trg`

Examples:

- `ddl_audit_trg`
- `logon_trg`



## Table

Plural<sup>1</sup> name of what is contained in the table (unless the table is designed to always hold one row only – then you should use a singular name).

Suffixed by `_eb` when protected by an editioning view.

Add a comment to the database dictionary for every table and every column in the table.

Optionally prefixed by a project abbreviation.

Examples:

- `employees`
- `departments`
- `countries_eb` - table interfaced by an editioning view named `countries`
- `sct_contracts`
- `sct_contract_lines`
- `sct_incentive_modules`

## Temporary Table (Global Temporary Table)

Naming as described for tables.

Optionally suffixed by `_tmp`

Optionally prefixed by a project abbreviation.

Examples:

- `employees_tmp`
- `contracts_tmp`

## Unique Key Constraint

Table name or table abbreviation followed by the role of the unique key constraint, a `_uk` and an optional number suffix.

Examples:

- `employees_name_uk`
- `departments_deptno_uk`
- `sct_contracts_uk`
- `sct_coli_uk`
- `sct_icmd_uk1`

## View

Plural<sup>1</sup> name of what is contained in the view. Optionally suffixed by an indicator identifying the object as a view (mostly used, when a 1:1 view layer lies above the table layer)

Editioning views are named like the original underlying table to avoid changing the existing application code when introducing edition based redefinition (EBR).

Add a comment to the database dictionary for every view and every column.

Optionally prefixed by a project abbreviation.

Examples:

- `active_orders`
- `orders_v` - a view to the orders table
- `countries` - an editioning view for table `countries_eb`

# Coding Style

## Formatting

### Rules

| Rule | Description                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------|
| 1    | Keywords are written uppercase, names are written in lowercase.                                                              |
| 2    | 3 space indentation <sup>2</sup> .                                                                                           |
| 3    | One command per line.                                                                                                        |
| 4    | Keywords <code>LOOP</code> , <code>ELSE</code> , <code>ELSIF</code> , <code>END IF</code> , <code>WHEN</code> on a new line. |
| 5    | Commas in front of separated elements.                                                                                       |
| 6    | Call parameters aligned, operators aligned, values aligned.                                                                  |
| 7    | SQL keywords are right aligned within a SQL command.                                                                         |
| 8    | Within a program unit only line comments <code>--</code> are used.                                                           |
| 9    | Brackets are used when needed or when helpful to clarify a construct.                                                        |

### Example

```

PROCEDURE set_salary(in_employee_id IN employees.employee_id%TYPE) IS
  CURSOR c_employees(p_employee_id IN employees.employee_id%TYPE) IS
    SELECT last_name
           ,first_name
           ,salary
    FROM employees
   WHERE employee_id = p_employee_id
   ORDER BY last_name
           ,first_name;

  r_employee      c_employees%ROWTYPE;
  l_new_salary    employees.salary%TYPE;
BEGIN
  OPEN c_employees(p_employee_id => in_employee_id);
  FETCH c_employees INTO r_employee;
  CLOSE c_employees;

  new_salary (in_employee_id => in_employee_id
             ,out_salary      => l_new_salary);

  -- Check whether salary has changed
  IF r_employee.salary <> l_new_salary THEN
    UPDATE employees
      SET salary = l_new_salary
    WHERE employee_id = in_employee_id;
  END IF;
END set_salary;

```

# Code Commenting

## Conventions

Inside a program unit only use the line commenting technique `--` unless you temporarily deactivate code sections for testing.

To comment the source code for later document generation, comments like `/** ... */` are used. Within these documentation comments, tags may be used to define the documentation structure.

Tools like ORACLE SQL Developer or PL/SQL Developer include documentation functionality based on a javadoc-like tagging.

## Commenting Tags

| Tag                 | Meaning                                                 | Example                                        |
|---------------------|---------------------------------------------------------|------------------------------------------------|
| <code>param</code>  | Description of a parameter.                             | <code>@param in_string input string</code>     |
| <code>return</code> | Description of the return value of a function.          | <code>@return result of the calculation</code> |
| <code>throws</code> | Describe errors that may be raised by the program unit. | <code>@throws NO_DATA_FOUND</code>             |

## Example

This is an example using the documentation capabilities of SQL Developer.

```
/**
Check whether we passed a valid sql name

@param   in_name  string to be checked
@return  in_name  if the string represents a valid sql name
@throws  ORA-44003: invalid SQL name

<b>Call Example:</b>
<pre>
    SELECT TVDAssert.valid_sql_name('TEST') from dual;
    SELECT TVDAssert.valid_sql_name('123') from dual
</pre>
*/
```

# Language Usage

## General

G-1010: Try to label your sub blocks.

 **Minor**

Maintainability

### Reason

It's a good alternative for comments to indicate the start and end of a named processing.

### Example (bad)

```
BEGIN
  BEGIN
    NULL;
  END;

  BEGIN
    NULL;
  END;
END;
/
```

### Example (good)

```
BEGIN
  <<prepare_data>>
  BEGIN
    NULL;
  END prepare_data;

  <<process_data>>
  BEGIN
    NULL;
  END process_data;
END good;
/
```

## G-1020: Always have a matching loop or block label.



Minor

Maintainability

### Reason

Use a label directly in front of loops and nested anonymous blocks:

- To give a name to that portion of code and thereby self-document what it is doing.
- So that you can repeat that name with the END statement of that block or loop.

### Example (bad)

```
DECLARE
  i INTEGER;
  co_min_value CONSTANT INTEGER := 1;
  co_max_value CONSTANT INTEGER := 10;
  co_increment CONSTANT INTEGER := 1;
BEGIN
  <<prepare_data>>
  BEGIN
    NULL;
  END;

  <<process_data>>
  BEGIN
    NULL;
  END;

  i := co_min_value;
  <<while_loop>>
  WHILE (i <= co_max_value)
  LOOP
    i := i + co_increment;
  END LOOP;

  <<basic_loop>>
  LOOP
    EXIT basic_loop;
  END LOOP;

  <<for_loop>>
  FOR i IN co_min_value..co_max_value
  LOOP
    sys.dbms_output.put_line(i);
  END LOOP;
END;
/
```

### Example (good)

```

DECLARE
    i INTEGER;
    co_min_value CONSTANT INTEGER := 1;
    co_max_value CONSTANT INTEGER := 10;
    co_increment CONSTANT INTEGER := 1;
BEGIN
    <<prepare_data>>
    BEGIN
        NULL;
    END prepare_data;

    <<process_data>>
    BEGIN
        NULL;
    END process_data;

    i := co_min_value;
    <<while_loop>>
    WHILE (i <= co_max_value)
    LOOP
        i := i + co_increment;
    END LOOP while_loop;

    <<basic_loop>>
    LOOP
        EXIT basic_loop;
    END LOOP basic_loop;

    <<for_loop>>
    FOR i IN co_min_value..co_max_value
    LOOP
        sys.dbms_output.put_line(i);
    END LOOP for_loop;
END;
/

```



## G-1030: Avoid defining variables that are not used.



Minor

Efficiency, Maintainability

### Reason

Unused variables decrease the maintainability and readability of your code.

### Example (bad)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_last_name employees.last_name%TYPE;
    l_first_name employees.first_name%TYPE;
    co_department_id CONSTANT departments.department_id%TYPE := 10;
    e_good EXCEPTION;
  BEGIN
    SELECT e.last_name
      INTO l_last_name
    FROM employees e
    WHERE e.department_id = co_department_id;
  EXCEPTION
    WHEN no_data_found THEN NULL; -- handle_no_data_found;
    WHEN too_many_rows THEN null; -- handle_too_many_rows;
  END my_proc;
END my_package;
/
```

### Example (good)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_last_name employees.last_name%TYPE;
    co_department_id CONSTANT departments.department_id%TYPE := 10;
    e_good EXCEPTION;
  BEGIN
    SELECT e.last_name
      INTO l_last_name
    FROM employees e
    WHERE e.department_id = co_department_id;

    RAISE e_good;
  EXCEPTION
    WHEN no_data_found THEN NULL; -- handle_no_data_found;
    WHEN too_many_rows THEN null; -- handle_too_many_rows;
  END my_proc;
END my_package;
/
```

## G-1040: Avoid dead code.



Minor

Maintainability

### Reason

Any part of your code, which is no longer used or cannot be reached, should be eliminated from your programs to simplify the code.

### Example (bad)

```
DECLARE
    co_dept_purchasing CONSTANT departments.department_id%TYPE := 30;
BEGIN
    IF 2=3 THEN
        NULL; -- some dead code here
    END IF;

    NULL; -- some enabled code here

    <<my_loop>>
    LOOP
        EXIT my_loop;
        NULL; -- some dead code here
    END LOOP my_loop;

    NULL; -- some other enabled code here

    CASE
        WHEN 1 = 1 AND 'x' = 'y' THEN
            NULL; -- some dead code here
        ELSE
            NULL; -- some further enabled code here
    END CASE;

    <<my_loop2>>
    FOR r_emp IN (SELECT last_name
                   FROM employees
                   WHERE department_id = co_dept_purchasing
                      OR commission_pct IS NOT NULL
                      AND 5=6)
        -- "OR commission_pct IS NOT NULL" is dead code
    LOOP
        SYS.dbms_output.put_line(r_emp.last_name);
    END LOOP my_loop2;

    RETURN;
    NULL; -- some dead code here
END;
/
```

### Example (good)

```

DECLARE
    co_dept_admin CONSTANT dept.deptno%TYPE := 10;
BEGIN
    NULL; -- some enabled code here
    NULL; -- some other enabled code here
    NULL; -- some further enabled code here

    <<my_loop2>>
    FOR r_emp IN (SELECT last_name
                   FROM employees
                   WHERE department_id = co_dept_admin
                     OR commission_pct IS NOT NULL)
    LOOP
        sys.dbms_output.put_line(r_emp.last_name);
    END LOOP my_loop2;
END;
/

```

## G-1050: Avoid using literals in your code.



Minor

Changeability

### Reason

Literals are often used more than once in your code. Having them defined as a constant reduces typos in your code and improves the maintainability.

All constants should be collated in just one package used as a library. If these constants should be used in SQL too it is good practice to write a deterministic package function for every constant.

### Example (bad)

```
DECLARE
    l_job employees.job_id%TYPE;
BEGIN
    SELECT e.job_id
        INTO l_job
        FROM employees e
        WHERE e.manager_id IS NULL;

    IF l_job = 'AD_PRES' THEN
        NULL;
    END IF;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        NULL; -- handle_no_data_found;
    WHEN TOO_MANY_ROWS THEN
        NULL; -- handle_too_many_rows;
END;
/
```

### Example (good)

```
CREATE OR REPLACE PACKAGE constants_up IS
    co_president CONSTANT employees.job_id%TYPE := 'AD_PRES';
END constants_up;
/

DECLARE
    l_job employees.job_id%TYPE;
BEGIN
    SELECT e.job_id
        INTO l_job
        FROM employees e
        WHERE e.manager_id IS NULL;

    IF l_job = constants_up.co_president THEN
        NULL;
    END IF;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        NULL; -- handle_no_data_found;
    WHEN TOO_MANY_ROWS THEN
        NULL; -- handle_too_many_rows;
END;
/
```

## G-1060: Avoid storing ROWIDs or UROWIDs in database tables.

### Major

Reliability

#### Reason

It is an extremely dangerous practice to store ROWIDs in a table, except for some very limited scenarios of runtime duration. Any manually explicit or system generated implicit table reorganization will reassign the row's ROWID and break the data consistency.

Instead of using ROWID for later reference to the original row one should use the primary key column(s).

#### Example (bad)

```
BEGIN
  INSERT INTO employees_log (employee_id
                           ,last_name
                           ,first_name
                           ,rid)

  SELECT employee_id
         ,last_name
         ,first_name
         ,ROWID
  FROM employees;
END;
/
```

#### Example (good)

```
BEGIN
  INSERT INTO employees_log (employee_id
                           ,last_name
                           ,first_name)

  SELECT employee_id
         ,last_name
         ,first_name
  FROM employees;
END;
/
```

## G-1070: Avoid nesting comment blocks.



Maintainability

### Reason

Having an end-of-comment within a block comment will end that block-comment. This does not only influence your code but is also very hard to read.

### Example (bad)

```
BEGIN
  /* comment one -- nested comment two */
  NULL;
  -- comment three /* nested comment four */
  NULL;
END;
/
```

### Example (good)

```
BEGIN
  /* comment one, comment two */
  NULL;
  -- comment three, comment four
  NULL;
END;
/
```

# Variables & Types

## General

**G-2110: Try to use anchored declarations for variables, constants and types.**

### Major

Maintainability, Reliability

#### REASON

Changing the size of the database column last\_name in the employees table from VARCHAR2(20) to VARCHAR2(30) will result in an error within your code whenever a value larger than the hard coded size is read from the table. This can be avoided using anchored declarations.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_last_name VARCHAR2(20 CHAR);
    co_first_row CONSTANT INTEGER := 1;
  BEGIN
    SELECT e.last_name
      INTO l_last_name
    FROM employees e
    WHERE rownum = co_first_row;
  EXCEPTION
    WHEN no_data_found THEN NULL; -- handle no_data_found
    WHEN too_many_rows THEN NULL; -- handle too_many_rows (impossible)
  END my_proc;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_last_name employees.last_name%TYPE;
    co_first_row CONSTANT INTEGER := 1;
  BEGIN
    SELECT e.last_name
      INTO l_last_name
    FROM employees e
    WHERE rownum = co_first_row;
  EXCEPTION
    WHEN no_data_found THEN NULL; -- handle no_data_found
    WHEN too_many_rows THEN NULL; -- handle too_many_rows (impossible)
  END my_proc;
END my_package;
/
```

## G-2120: Try to have a single location to define your types.



Minor

Changeability

### REASON

Single point of change when changing the data type. No need to argue where to define types or where to look for existing definitions.

A single location could be either a type specification package or the database (database-defined types).

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    SUBTYPE big_string_type IS VARCHAR2(1000 CHAR);
    l_note big_string_type;
  BEGIN
    l_note := some_function();
  END my_proc;
END my_package;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE types_up IS
  SUBTYPE big_string_type IS VARCHAR2(1000 CHAR);
END types_up;
/

CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_note types_up.big_string_type;
  BEGIN
    l_note := some_function();
  END my_proc;
END my_package;
/
```



G-2130: Try to use subtypes for constructs used often in your code.

 Minor

Changeability

REASON

Single point of change when changing the data type.

Your code will be easier to read as the usage of a variable/constant may be derived from its definition.

EXAMPLES OF POSSIBLE SUBTYPE DEFINITIONS

| Type             | Usage                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------|
| ora_name_type    | Object corresponding to the ORACLE naming conventions (table, variable, column, package, etc.). |
| max_vc2_type     | String variable with maximal VARCHAR2 size.                                                     |
| array_index_type | Best fitting data type for array navigation.                                                    |
| id_type          | Data type used for all primary key (id) columns.                                                |

EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_note VARCHAR2(1000 CHAR);
  BEGIN
    l_note := some_function();
  END my_proc;
END my_package;
/
```

EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE types_up IS
  SUBTYPE big_string_type IS VARCHAR2(1000 CHAR);
END types_up;
/

CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_proc IS
    l_note types_up.big_string_type;
  BEGIN
    l_note := some_function();
  END my_proc;
END my_package;
/
```

## G-2140: Never initialize variables with NULL.



Minor

Maintainability

### REASON

Variables are initialized to NULL by default.

### EXAMPLE (BAD)

```
DECLARE
    l_note big_string_type := NULL;
BEGIN
    sys.dbms_output.put_line(l_note);
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
    l_note big_string_type;
BEGIN
    sys.dbms_output.put_line(l_note);
END;
/
```

## G-2150: Avoid comparisons with NULL value, consider using IS [NOT] NULL.

### Blocker

Portability, Reliability

#### REASON

The NULL value can cause confusion both from the standpoint of code review and code execution. You must always use the `IS NULL` or `IS NOT NULL` syntax when you need to check if a value is or is not `NULL`.

#### EXAMPLE (BAD)

```
DECLARE
    l_value INTEGER;
BEGIN
    IF l_value = NULL THEN
        NULL;
    END IF;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    l_value INTEGER;
BEGIN
    IF l_value IS NULL THEN
        NULL;
    END IF;
END;
/
```

## G-2160: Avoid initializing variables using functions in the declaration section.

### Critical

Reliability

#### REASON

If your initialization fails, you will not be able to handle the error in your exceptions block.

#### EXAMPLE (BAD)

```
DECLARE
    co_department_id CONSTANT INTEGER := 100;
    l_department_name departments.department_name%TYPE :=
        department_api.name_by_id(in_id => co_department_id);
BEGIN
    sys.dbms_output.put_line(l_department_name);
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    co_department_id CONSTANT INTEGER := 100;
    co_unkown_name CONSTANT departments.department_name%TYPE := 'unknown';
    l_department_name departments.department_name%TYPE;
BEGIN
    <<init>>
    BEGIN
        l_department_name := department_api.name_by_id(in_id => co_department_id);
    EXCEPTION
        WHEN value_error THEN
            l_department_name := co_unkown_name;
    END init;

    sys.dbms_output.put_line(l_department_name);
END;
/
```

## G-2170: Never overload variables.

### Major

Reliability

#### REASON

The readability of your code will be higher when you do not overload variables.

#### EXAMPLE (BAD)

```
BEGIN
  <<main>>
  DECLARE
    co_main CONSTANT user_objects.object_name%TYPE := 'test_main';
    co_sub  CONSTANT user_objects.object_name%TYPE := 'test_sub';
    co_sep  CONSTANT user_objects.object_name%TYPE := ' - ';
    l_variable user_objects.object_name%TYPE := co_main;
  BEGIN
    <<sub>>
    DECLARE
      l_variable user_objects.object_name%TYPE := co_sub;
    BEGIN
      sys.dbms_output.put_line(l_variable || co_sep || main.l_variable);
    END sub;
  END main;
END;
/
```

#### EXAMPLE (GOOD)

```
BEGIN
  <<main>>
  DECLARE
    co_main CONSTANT user_objects.object_name%TYPE := 'test_main';
    co_sub  CONSTANT user_objects.object_name%TYPE := 'test_sub';
    co_sep  CONSTANT user_objects.object_name%TYPE := ' - ';
    l_main_variable user_objects.object_name%TYPE := co_main;
  BEGIN
    <<sub>>
    DECLARE
      l_sub_variable user_objects.object_name%TYPE := co_sub;
    BEGIN
      sys.dbms_output.put_line(l_sub_variable || co_sep || l_main_variable);
    END sub;
  END main;
END;
/
```

## G-2180: Never use quoted identifiers.

### Major

Maintainability

#### REASON

Quoted identifiers make your code hard to read and maintain.

#### EXAMPLE (BAD)

```
DECLARE
    "sal+comm" INTEGER;
    "my constant" CONSTANT INTEGER := 1;
    "my exception" EXCEPTION;
BEGIN
    "sal+comm" := "my constant";
EXCEPTION
    WHEN "my exception" THEN
        NULL;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    l_sal_comm    INTEGER;
    co_my_constant CONSTANT INTEGER := 1;
    e_my_exception EXCEPTION;
BEGIN
    l_sal_comm := co_my_constant;
EXCEPTION
    WHEN e_my_exception THEN
        NULL;
END;
/
```

## G-2185: Avoid using overly short names for explicitly or implicitly declared identifiers.



Minor

Maintainability

### REASON

You should ensure that the name you have chosen well defines its purpose and usage. While you can save a few keystrokes typing very short names, the resulting code is obscure and hard for anyone besides the author to understand.

### EXAMPLE (BAD)

```
DECLARE
  i INTEGER;
  c CONSTANT INTEGER := 1;
  e EXCEPTION;
BEGIN
  i := c;
EXCEPTION
  WHEN e THEN
    NULL;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  l_sal_comm      INTEGER;
  co_my_constant CONSTANT INTEGER := 1;
  e_my_exception  EXCEPTION;
BEGIN
  l_sal_comm := co_my_constant;
EXCEPTION
  WHEN e_my_exception THEN
    NULL;
END;
/
```

## G-2190: Avoid using ROWID or UROWID.

### Major

Portability, Reliability

#### REASON

Be careful about your use of Oracle-specific data types like `ROWID` and `UROWID`. They might offer a slight improvement in performance over other means of identifying a single row (primary key or unique index value), but that is by no means guaranteed.

Use of `ROWID` or `UROWID` means that your SQL statement will not be portable to other SQL databases. Many developers are also not familiar with these data types, which can make the code harder to maintain.

#### EXAMPLE (BAD)

```
DECLARE
    l_department_name departments.department_name%TYPE;
    l_rowid ROWID;
BEGIN
    UPDATE departments
        SET department_name = l_department_name
        WHERE ROWID = l_rowid;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    l_department_name departments.department_name%TYPE;
    l_department_id departments.department_id%TYPE;
BEGIN
    UPDATE departments
        SET department_name = l_department_name
        WHERE department_id = l_department_id;
END;
/
```



## Numeric Data Types

### G-2210: Avoid declaring NUMBER variables, constants or subtypes with no precision.

 Minor

Efficiency

#### REASON

If you do not specify precision `NUMBER` is defaulted to 38 or the maximum supported by your system, whichever is less. You may well need all this precision, but if you know you do not, you should specify whatever matches your needs.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_small_increase CONSTANT NUMBER := 0.1;

    FUNCTION small_increase RETURN NUMBER IS
    BEGIN
        RETURN co_small_increase;
    END small_increase;
END constants_up;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_small_increase CONSTANT NUMBER(5,1) := 0.1;

    FUNCTION small_increase RETURN NUMBER IS
    BEGIN
        RETURN co_small_increase;
    END small_increase;
END constants_up;
/
```

## G-2220: Try to use PLS\_INTEGER instead of NUMBER for arithmetic operations with integer values.

 Minor

Efficiency

### REASON

PLS\_INTEGER having a length of -2,147,483,648 to 2,147,483,647, on a 32bit system.

There are many reasons to use PLS\_INTEGER instead of NUMBER :

- PLS\_INTEGER uses less memory
- PLS\_INTEGER uses machine arithmetic, which is up to three times faster than library arithmetic, which is used by NUMBER .

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_big_increase CONSTANT NUMBER(5,0) := 1;

    FUNCTION big_increase RETURN NUMBER IS
    BEGIN
        RETURN co_big_increase;
    END big_increase;
END constants_up;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_big_increase CONSTANT PLS_INTEGER := 1;

    FUNCTION big_increase RETURN PLS_INTEGER IS
    BEGIN
        RETURN co_big_increase;
    END big_increase;
END constants_up;
/
```

## G-2230: Try to use SIMPLE\_INTEGER datatype when appropriate.



Minor

Efficiency

### RESTRICTION

ORACLE 11g or later

### REASON

`SIMPLE_INTEGER` does no checks on numeric overflow, which results in better performance compared to the other numeric datatypes.

With ORACLE 11g, the new data type `SIMPLE_INTEGER` has been introduced. It is a sub-type of `PLS_INTEGER` and covers the same range. The basic difference is that `SIMPLE_INTEGER` is always `NOT NULL`. When the value of the declared variable is never going to be null then you can declare it as `SIMPLE_INTEGER`. Another major difference is that you will never face a numeric overflow using `SIMPLE_INTEGER` as this data type wraps around without giving any error.

`SIMPLE_INTEGER` data type gives major performance boost over `PLS_INTEGER` when code is compiled in `NATIVE` mode, because arithmetic operations on `SIMPLE_INTEGER` type are performed directly at the hardware level.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_big_increase CONSTANT NUMBER(5,0) := 1;

    FUNCTION big_increase RETURN NUMBER IS
    BEGIN
        RETURN co_big_increase;
    END big_increase;
END constants_up;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_big_increase CONSTANT SIMPLE_INTEGER := 1;

    FUNCTION big_increase RETURN SIMPLE_INTEGER IS
    BEGIN
        RETURN co_big_increase;
    END big_increase;
END constants_up;
/
```

## Character Data Types

### G-2310: Avoid using CHAR data type.

 **Major**

Reliability

#### REASON

CHAR is a fixed length data type, which should only be used when appropriate. CHAR columns/variables are always filled to its specified lengths; this may lead to unwanted side effects and undesired results.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE types_up
IS
    SUBTYPE description_type IS CHAR(200);
END types_up;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE types_up
IS
    SUBTYPE description_type IS VARCHAR2(200 CHAR);
END types_up;
/
```

## G-2320: Avoid using VARCHAR data type.

### Major

Portability

#### REASON

Do not use the `VARCHAR` data type. Use the `VARCHAR2` data type instead. Although the `VARCHAR` data type is currently synonymous with `VARCHAR2`, the `VARCHAR` data type is scheduled to be redefined as a separate data type used for variable-length character strings compared with different comparison semantics.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE types_up IS
    SUBTYPE description_type IS VARCHAR(200);
END types_up;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE types_up IS
    SUBTYPE description_type IS VARCHAR2(200 CHAR);
END types_up;
/
```

## G-2330: Never use zero-length strings to substitute NULL.

### Major

Portability

#### REASON

Today zero-length strings and `NULL` are currently handled identical by ORACLE. There is no guarantee that this will still be the case in future releases, therefore if you mean `NULL` use `NULL`.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS
    co_null_string CONSTANT VARCHAR2(1) := '';

    FUNCTION null_string RETURN VARCHAR2 IS
    BEGIN
        RETURN co_null_string;
    END null_string;
END constants_up;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY constants_up IS

    FUNCTION empty_string RETURN VARCHAR2 IS
    BEGIN
        RETURN NULL;
    END empty_string;
END constants_up;
/
```

**G-2340: Always define your VARCHAR2 variables using CHAR SEMANTIC (if not defined anchored).**

 **Minor**

Reliability

**REASON**

Changes to the `NLS_LENGTH_SEMANTIC` will only be picked up by your code after a recompilation.

In a multibyte environment a `VARCHAR2(10)` definition may not necessarily hold 10 characters, when multibyte characters a part of the value that should be stored unless the definition was done using the char semantic.

**EXAMPLE (BAD)**

```
CREATE OR REPLACE PACKAGE types_up IS
  SUBTYPE description_type IS VARCHAR2(200);
END types_up;
/
```

**EXAMPLE (GOOD)**

```
CREATE OR REPLACE PACKAGE types_up IS
  SUBTYPE description_type IS VARCHAR2(200 CHAR);
END types_up;
/
```

## Boolean Data Types

**G-2410: Try to use boolean data type for values with dual meaning.**



Minor

Maintainability

### REASON

The use of TRUE and FALSE clarifies that this is a boolean value and makes the code easier to read.

### EXAMPLE (BAD)

```
DECLARE
  co_newFile CONSTANT PLS_INTEGER := 1000;
  co_oldFile  CONSTANT PLS_INTEGER := 500;
  l_bigger    PLS_INTEGER;
BEGIN
  IF co_newFile < co_oldFile THEN
    l_bigger := constants_up.co_numeric_true;
  ELSE
    l_bigger := constants_up.co_numeric_false;
  END IF;
END;
/
```

### EXAMPLE (BETTER)

```
DECLARE
  co_newFile CONSTANT PLS_INTEGER := 1000;
  co_oldFile  CONSTANT PLS_INTEGER := 500;
  l_bigger    BOOLEAN;
BEGIN
  IF co_newFile < co_oldFile THEN
    l_bigger := TRUE;
  ELSE
    l_bigger := FALSE;
  END IF;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  co_newFile CONSTANT PLS_INTEGER := 1000;
  co_oldFile  CONSTANT PLS_INTEGER := 500;
  l_bigger    BOOLEAN;
BEGIN
  l_bigger := NVL(co_newFile < co_oldFile, FALSE);
END;
/
```



## Large Objects

### G-2510: Avoid using the LONG and LONG RAW data types.

#### Major

Portability

#### REASON

LONG and LONG RAW data types have been deprecated by ORACLE since version 8i - support might be discontinued in future ORACLE releases.

There are many constraints to LONG datatypes in comparison to the LOB types.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE example_package IS
    g_long LONG;
    g_raw LONG RAW;

    PROCEDURE do_something;
END example_package;
/

CREATE OR REPLACE PACKAGE BODY example_package IS
    PROCEDURE do_something IS
    BEGIN
        NULL;
    END do_something;
END example_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE example_package IS
    PROCEDURE do_something;
END example_package;
/

CREATE OR REPLACE PACKAGE BODY example_package IS
    g_long CLOB;
    g_raw BLOB;

    PROCEDURE do_something IS
    BEGIN
        NULL;
    END do_something;
END example_package;
/
```

# DML & SQL

## General

**G-3110: Always specify the target columns when coding an insert statement.**

 **Major**

Maintainability, Reliability

### REASON

Data structures often change. Having the target columns in your insert statements will lead to change-resistant code.

### EXAMPLE (BAD)

```
INSERT INTO departments
VALUES (departments_seq.nextval
      , 'Support'
      , 100
      , 10);
```

### EXAMPLE (GOOD)

```
INSERT INTO departments (department_id
                        , department_name
                        , manager_id
                        , location_id)
VALUES (departments_seq.nextval
      , 'Support'
      , 100
      , 10);
```

## G-3120: Always use table aliases when your SQL statement involves more than one source.

### Major

Maintainability

#### REASON

It is more human readable to use aliases instead of writing columns with no table information.

Especially when using subqueries the omission of table aliases may end in unexpected behavior and result.

#### EXAMPLE (BAD)

```
SELECT last_name
       ,first_name
       ,department_name
FROM   employees
       JOIN departments USING (department_id)
WHERE  EXTRACT(MONTH FROM hire_date) = EXTRACT(MONTH FROM SYSDATE);
```

#### EXAMPLE (BETTER)

```
SELECT e.last_name
       ,e.first_name
       ,d.department_name
FROM   employees e
       JOIN departments d ON (e.department_id = d.department_id)
WHERE  EXTRACT(MONTH FROM e.hire_date) = EXTRACT(MONTH FROM SYSDATE);
```

#### EXAMPLE (GOOD)

Using meaningful aliases improves the readability of your code.

```
SELECT emp.last_name
       ,emp.first_name
       ,dept.department_name
FROM   employees emp
       JOIN departments dept ON (emp.department_id = dept.department_id)
WHERE  EXTRACT(MONTH FROM emp.hire_date) = EXTRACT(MONTH FROM SYSDATE);
```

#### EXAMPLE SUBQUERY (BAD)

If the `jobs` table has no `employee_id` column and `employees` has one this query will not raise an error but return all rows of the `employees` table as a subquery is allowed to access columns of all its parent tables - this construct is known as correlated subquery.

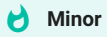
```
SELECT last_name
       ,first_name
FROM   employees
WHERE  employee_id IN (SELECT employee_id
                      FROM   jobs
                      WHERE  job_title like '%Manager%');
```

#### EXAMPLE SUBQUERY (GOOD)

If the `jobs` table has no `employee_id` column this query will return an error due to the directive (given by adding the table alias to the column) to read the `employee_id` column from the `jobs` table.

```
SELECT emp.last_name
       ,emp.first_name
  FROM employees emp
 WHERE emp.employee_id IN (SELECT j.employee_id
                           FROM jobs j
                           WHERE j.job_title like '%Manager%');
```

### G-3130: Try to use ANSI SQL-92 join syntax.



Minor

Maintainability, Portability

#### REASON

ANSI SQL-92 join syntax supports the full outer join. A further advantage of the ANSI SQL-92 join syntax is the separation of the join condition from the query filters.

#### EXAMPLE (BAD)

```
SELECT e.employee_id
       ,e.last_name
       ,e.first_name
       ,d.department_name
FROM   employees e
       ,departments d
WHERE  e.department_id = d.department_id
       AND EXTRACT(MONTH FROM e.hire_date) = EXTRACT(MONTH FROM SYSDATE);
```

#### EXAMPLE (GOOD)

```
SELECT emp.employee_id
       ,emp.last_name
       ,emp.first_name
       ,dept.department_name
FROM   employees emp
       JOIN departments dept ON dept.department_id = emp.department_id
WHERE  EXTRACT(MONTH FROM emp.hire_date) = EXTRACT(MONTH FROM SYSDATE);
```

## G-3140: Try to use anchored records as targets for your cursors.

### Major

Maintainability, Reliability

#### REASON

Using cursor-anchored records as targets for your cursors results enables the possibility of changing the structure of the cursor without regard to the target structure.

#### EXAMPLE (BAD)

```
DECLARE
  CURSOR c_employees IS
    SELECT employee_id, first_name, last_name
      FROM employees;
  l_employee_id employees.employee_id%TYPE;
  l_first_name  employees.first_name%TYPE;
  l_last_name   employees.last_name%TYPE;
BEGIN
  OPEN c_employees;
  FETCH c_employees INTO l_employee_id, l_first_name, l_last_name;
  <<process_employees>>
  WHILE c_employees%FOUND
  LOOP
    -- do something with the data
    FETCH c_employees INTO l_employee_id, l_first_name, l_last_name;
  END LOOP process_employees;
  CLOSE c_employees;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
  CURSOR c_employees IS
    SELECT employee_id, first_name, last_name
      FROM employees;
  r_employee c_employees%ROWTYPE;
BEGIN
  OPEN c_employees;
  FETCH c_employees INTO r_employee;
  <<process_employees>>
  WHILE c_employees%FOUND
  LOOP
    -- do something with the data
    FETCH c_employees INTO r_employee;
  END LOOP process_employees;
  CLOSE c_employees;
END;
/
```

## G-3150: Try to use identity columns for surrogate keys.



Minor

Maintainability, Reliability

### RESTRICTION

ORACLE 12c

### REASON

An identity column is a surrogate key by design – there is no reason why we should not take advantage of this natural implementation when the keys are generated on database level. Using identity column (and therefore assigning sequences as default values on columns) has a huge performance advantage over a trigger solution.

### EXAMPLE (BAD)

```
CREATE TABLE locations (
  location_id      NUMBER(10)      NOT NULL
,location_name    VARCHAR2(60 CHAR) NOT NULL
,city             VARCHAR2(30 CHAR) NOT NULL
,CONSTRAINT locations_pk PRIMARY KEY (location_id)
)
/

CREATE SEQUENCE location_seq START WITH 1 CACHE 20
/

CREATE OR REPLACE TRIGGER location_br_i
  BEFORE INSERT ON LOCATIONS
  FOR EACH ROW
BEGIN
  :new.location_id := location_seq.nextval;
END;
/
```

### EXAMPLE (GOOD)

```
CREATE TABLE locations (
  location_id      NUMBER(10) GENERATED ALWAYS AS IDENTITY
,location_name    VARCHAR2(60 CHAR) NOT NULL
,city             VARCHAR2(30 CHAR) NOT NULL
,CONSTRAINT locations_pk PRIMARY KEY (location_id))
/
```

`GENERATED ALWAYS AS IDENTITY` ensures that the `location_id` is populated by a sequence. It is not possible to override the behavior in the application.

However, if you use a framework that produces an `INSERT` statement including the surrogate key column, and you cannot change this behavior, then you have to use the `GENERATED BY DEFAULT ON NULL AS IDENTITY` option. This has the downside that the application may pass a value, which might lead to an immediate or delayed `ORA-00001: unique constraint violated` error.

## G-3160: Avoid visible virtual columns.

### Major

Maintainability, Reliability

#### RESTRICTION

ORACLE 12c

#### REASON

In contrast to visible columns, invisible columns are not part of a record defined using `%ROWTYPE` construct. This is helpful as a virtual column may not be programmatically populated. If your virtual column is visible you have to manually define the record types used in API packages to be able to exclude them from being part of the record definition.

Invisible columns may be accessed by explicitly adding them to the column list in a SELECT statement.

#### EXAMPLE (BAD)

```
ALTER TABLE employees
  ADD total_salary GENERATED ALWAYS AS (salary + NVL(commission_pct,0) * salary)
/

DECLARE
  r_employee employees%ROWTYPE;
  l_id employees.employee_id%TYPE := 107;
BEGIN
  r_employee := employee_api.employee_by_id(l_id);
  r_employee.salary := r_employee.salary * constants_up.small_increase();

  UPDATE employees
    SET ROW = r_employee
    WHERE employee_id = l_id;
END;
/

Error report -
ORA-54017: UPDATE operation disallowed ON virtual COLUMNS
ORA-06512: at line 9
```

#### EXAMPLE (GOOD)

```
ALTER TABLE employees
  ADD total_salary INVISIBLE GENERATED ALWAYS AS
    (salary + NVL(commission_pct,0) * salary)
/

DECLARE
  r_employee employees%ROWTYPE;
  co_id CONSTANT employees.employee_id%TYPE := 107;
BEGIN
  r_employee := employee_api.employee_by_id(co_id);
  r_employee.salary := r_employee.salary * constants_up.small_increase();

  UPDATE employees
    SET ROW = r_employee
    WHERE employee_id = co_id;
END;
/
```



**G-3170: Always use DEFAULT ON NULL declarations to assign default values to table columns if you refuse to store NULL values.**

 **Major**

Reliability

RESTRICTION

ORACLE 12c

REASON

Default values have been nullifiable until ORACLE 12c. Meaning any tool sending null as a value for a column having a default value bypassed the default value. Starting with ORACLE 12c default definitions may have an `ON NULL` definition in addition, which will assign the default value in case of a null value too.

EXAMPLE (BAD)

```
CREATE TABLE null_test (
  test_case          NUMBER(2) NOT NULL
  ,column_defaulted VARCHAR2(10 CHAR) DEFAULT 'Default')
/
INSERT INTO null_test(test_case, column_defaulted) VALUES (1,'Value');
INSERT INTO null_test(test_case, column_defaulted) VALUES (2,DEFAULT);
INSERT INTO null_test(test_case, column_defaulted) VALUES (3,NULL);

SELECT * FROM null_test;
```

| TEST_CASE | COLUMN_DEF |
|-----------|------------|
| 1         | Value      |
| 2         | Default    |
| 3         |            |

EXAMPLE (GOOD)

```
CREATE TABLE null_test (
  test_case          NUMBER(2) NOT NULL
  ,column_defaulted VARCHAR2(10 CHAR) DEFAULT ON NULL 'Default')
/
INSERT INTO null_test(test_case, column_defaulted) VALUES (1,'Value');
INSERT INTO null_test(test_case, column_defaulted) VALUES (2,DEFAULT);
INSERT INTO null_test(test_case, column_defaulted) VALUES (3,NULL);

SELECT * FROM null_test;
```

| TEST_CASE | COLUMN_DEF |
|-----------|------------|
| 1         | Value      |
| 2         | Default    |
| 3         | Default    |

## G-3180: Always specify column names instead of positional references in ORDER BY clauses.

### Major

Changeability, Reliability

#### REASON

If you change your select list afterwards the ORDER BY will still work but order your rows differently, when not changing the positional number. Furthermore, it is not comfortable to the readers of the code, if they have to count the columns in the SELECT list to know the way the result is ordered.

#### EXAMPLE (BAD)

```
SELECT UPPER(first_name)
       ,last_name
       ,salary
       ,hire_date
FROM employees
ORDER BY 4,1,3;
```

#### EXAMPLE (GOOD)

```
SELECT upper(first_name) AS first_name
       ,last_name
       ,salary
       ,hire_date
FROM employees
ORDER BY hire_date
       ,first_name
       ,salary;
```

G-3190: Avoid using NATURAL JOIN.

 Major

Changeability, Reliability

REASON

A natural join joins tables on equally named columns. This may comfortably fit on first sight, but adding logging columns to a table (changed\_by, changed\_date) will result in inappropriate join conditions.

EXAMPLE (BAD)

```
SELECT department_name
       ,last_name
       ,first_name
  FROM employees NATURAL JOIN departments
 ORDER BY department_name
        ,last_name;
```

| DEPARTMENT_NAME | LAST_NAME | FIRST_NAME |
|-----------------|-----------|------------|
| Accounting      | Gietz     | William    |
| Executive       | De Haan   | Lex        |
| ...             |           |            |

```
ALTER TABLE departments ADD modified_at DATE DEFAULT ON NULL SYSDATE;
ALTER TABLE employees ADD modified_at DATE DEFAULT ON NULL SYSDATE;
```

```
SELECT department_name
       ,last_name
       ,first_name
  FROM employees NATURAL JOIN departments
 ORDER BY department_name
        ,last_name;
```

No data found

EXAMPLE (GOOD)

```
SELECT d.department_name
       ,e.last_name
       ,e.first_name
  FROM employees   e
 JOIN departments d ON (e.department_id = d.department_id)
 ORDER BY d.department_name
        ,e.last_name;
```

| DEPARTMENT_NAME | LAST_NAME | FIRST_NAME |
|-----------------|-----------|------------|
| Accounting      | Gietz     | William    |
| Executive       | De Haan   | Lex        |
| ...             |           |            |

## Bulk Operations

**G-3210: Always use BULK OPERATIONS (BULK COLLECT, FORALL) whenever you have to execute a DML statement for more than 4 times.**

### Major

Efficiency

#### REASON

Context switches between PL/SQL and SQL are extremely costly. BULK Operations reduce the number of switches by passing an array to the SQL engine, which is used to execute the given statements repeatedly.

(Depending on the PLSQL\_OPTIMIZE\_LEVEL parameter a conversion to BULK COLLECT will be done by the PL/SQL compiler automatically.)

#### EXAMPLE (BAD)

```
DECLARE
    t_employee_ids employee_api.t_employee_ids_type;
    co_increase CONSTANT employees.salary%type := 0.1;
    co_department_id CONSTANT departments.department_id%TYPE := 10;
BEGIN
    t_employee_ids := employee_api.employee_ids_by_department(
        id_in => co_department_id
    );
    <<process_employees>>
    FOR i IN 1..t_employee_ids.COUNT()
    LOOP
        UPDATE employees
            SET salary = salary + (salary * co_increase)
            WHERE employee_id = t_employee_ids(i);
    END LOOP process_employees;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    t_employee_ids employee_api.t_employee_ids_type;
    co_increase CONSTANT employees.salary%type := 0.1;
    co_department_id CONSTANT departments.department_id%TYPE := 10;
BEGIN
    t_employee_ids := employee_api.employee_ids_by_department(
        id_in => co_department_id
    );
    <<process_employees>>
    FORALL i IN 1..t_employee_ids.COUNT()
        UPDATE employees
            SET salary = salary + (salary * co_increase)
            WHERE employee_id = t_employee_ids(i);
END;
/
```

# Control Structures

## CURSOR

**G-4110: Always use %NOTFOUND instead of NOT %FOUND to check whether a cursor returned data.**



Minor

Maintainability

### REASON

The readability of your code will be higher when you avoid negative sentences.

### EXAMPLE (BAD)

```
DECLARE
  CURSOR c_employees IS
    SELECT last_name
           ,first_name
    FROM employees
    WHERE commission_pct IS NOT NULL;

  r_employee c_employees%ROWTYPE;
BEGIN
  OPEN c_employees;

  <<read_employees>>
  LOOP
    FETCH c_employees INTO r_employee;
    EXIT read_employees WHEN NOT c_employees%FOUND;
  END LOOP read_employees;

  CLOSE c_employees;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  CURSOR c_employees IS
    SELECT last_name
           ,first_name
    FROM employees
    WHERE commission_pct IS NOT NULL;

  r_employee c_employees%ROWTYPE;
BEGIN
  OPEN c_employees;

  <<read_employees>>
  LOOP
    FETCH c_employees INTO r_employee;
    EXIT read_employees WHEN c_employees%NOTFOUND;
  END LOOP read_employees;

  CLOSE c_employees;
END;
/
```

**G-4120: Avoid using %NOTFOUND directly after the FETCH when working with BULK OPERATIONS and LIMIT clause.**

 **Critical**

Reliability

**REASON**

%NOTFOUND is set to TRUE as soon as less than the number of rows defined by the LIMIT clause has been read.

**EXAMPLE (BAD)**

The employees table holds 107 rows. The example below will only show 100 rows as the cursor attribute NOTFOUND is set to true as soon as the number of rows to be fetched defined by the limit clause is not fulfilled anymore.

```
DECLARE
  CURSOR c_employees IS
    SELECT *
      FROM employees
     ORDER BY employee_id;

  TYPE t_employees_type IS TABLE OF c_employees%ROWTYPE;
  t_employees t_employees_type;
  co_bulk_size CONSTANT SIMPLE_INTEGER := 10;
BEGIN
  OPEN c_employees;

  <<process_employees>>
  LOOP
    FETCH c_employees BULK COLLECT INTO t_employees LIMIT co_bulk_size;
    EXIT process_employees WHEN c_employees%NOTFOUND;

    <<display_employees>>
    FOR i IN 1..t_employees.COUNT()
    LOOP
      sys.dbms_output.put_line(t_employees(i).last_name);
    END LOOP display_employees;
  END LOOP process_employees;

  CLOSE c_employees;
END;
/
```

**EXAMPLE (BETTER)**

This example will show all 107 rows but execute one fetch too much (12 instead of 11).

```

DECLARE
  CURSOR c_employees IS
    SELECT *
      FROM employees
     ORDER BY employee_id;

  TYPE t_employees_type IS TABLE OF c_employees%ROWTYPE;
  t_employees t_employees_type;
  co_bulk_size CONSTANT SIMPLE_INTEGER := 10;
BEGIN
  OPEN c_employees;

  <<process_employees>>
  LOOP
    FETCH c_employees BULK COLLECT INTO t_employees LIMIT co_bulk_size;
    EXIT process_employees WHEN t_employees.COUNT() = 0;
    <<display_employees>>
    FOR i IN 1..t_employees.COUNT()
      LOOP
        sys.dbms_output.put_line(t_employees(i).last_name);
      END LOOP display_employees;
    END LOOP process_employees;

  CLOSE c_employees;
END;
/

```

#### EXAMPLE (GOOD)

This example does the trick (11 fetches only to process all rows)

```

DECLARE
  CURSOR c_employees IS
    SELECT *
      FROM employees
     ORDER BY employee_id;

  TYPE t_employees_type IS TABLE OF c_employees%ROWTYPE;
  t_employees t_employees_type;
  co_bulk_size CONSTANT SIMPLE_INTEGER := 10;
BEGIN
  OPEN c_employees;

  <<process_employees>>
  LOOP
    FETCH c_employees BULK COLLECT INTO t_employees LIMIT co_bulk_size;
    <<display_employees>>
    FOR i IN 1..t_employees.COUNT()
      LOOP
        sys.dbms_output.put_line(t_employees(i).last_name);
      END LOOP display_employees;
    EXIT process_employees WHEN t_employees.COUNT() <> co_bulk_size;
  END LOOP process_employees;

  CLOSE c_employees;
END;
/

```

## G-4130: Always close locally opened cursors.

### Major

Efficiency, Reliability

#### REASON

Any cursors left open can consume additional memory space (i.e. SGA) within the database instance, potentially in both the shared and private SQL pools. Furthermore, failure to explicitly close cursors may also cause the owning session to exceed its maximum limit of open cursors (as specified by the `OPEN_CURSORS` database initialization parameter), potentially resulting in the Oracle error of "ORA-01000: maximum open cursors exceeded".

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY employee_api AS
  FUNCTION department_salary (in_dept_id IN departments.department_id%TYPE)
    RETURN NUMBER IS
    CURSOR c_department_salary(p_dept_id IN departments.department_id%TYPE) IS
      SELECT sum(salary) AS sum_salary
      FROM employees
      WHERE department_id = p_dept_id;
    r_department_salary c_department_salary%rowtype;
  BEGIN
    OPEN c_department_salary(p_dept_id => in_dept_id);
    FETCH c_department_salary INTO r_department_salary;

    RETURN r_department_salary.sum_salary;
  END department_salary;
END employee_api;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY employee_api AS
  FUNCTION department_salary (in_dept_id IN departments.department_id%TYPE)
    RETURN NUMBER IS
    CURSOR c_department_salary(p_dept_id IN departments.department_id%TYPE) IS
      SELECT SUM(salary) AS sum_salary
      FROM employees
      WHERE department_id = p_dept_id;
    r_department_salary c_department_salary%rowtype;
  BEGIN
    OPEN c_department_salary(p_dept_id => in_dept_id);
    FETCH c_department_salary INTO r_department_salary;
    CLOSE c_department_salary;
    RETURN r_department_salary.sum_salary;
  END department_salary;
END employee_api;
/
```



## G-4140: Avoid executing any statements between a SQL operation and the usage of an implicit cursor attribute.

### Major

Reliability

#### REASON

Oracle provides a variety of cursor attributes (like `%FOUND` and `%ROWCOUNT`) that can be used to obtain information about the status of a cursor, either implicit or explicit.

You should avoid inserting any statements between the cursor operation and the use of an attribute against that cursor. Interposing such a statement can affect the value returned by the attribute, thereby potentially corrupting the logic of your program.

In the following example, a procedure call is inserted between the DELETE statement and a check for the value of `SQL%ROWCOUNT`, which returns the number of rows modified by that last SQL statement executed in the session. If this procedure includes a `COMMIT` / `ROLLBACK` or another implicit cursor the value of `SQL%ROWCOUNT` is affected.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY employee_api AS
    co_one CONSTANT SIMPLE_INTEGER := 1;

    PROCEDURE process_dept(in_dept_id IN departments.department_id%TYPE) IS
    BEGIN
        NULL;
    END process_dept;

    PROCEDURE remove_employee (in_employee_id IN employees.employee_id%TYPE) IS
        l_dept_id      employees.department_id%TYPE;
    BEGIN
        DELETE FROM employees
        WHERE employee_id = in_employee_id
        RETURNING department_id INTO l_dept_id;

        process_dept(in_dept_id => l_dept_id);

        IF SQL%ROWCOUNT > co_one THEN
            -- too many rows deleted.
            ROLLBACK;
        END IF;
    END remove_employee;
END employee_api;
/
```

#### EXAMPLE (GOOD)

```

CREATE OR REPLACE PACKAGE BODY employee_api AS
    co_one CONSTANT SIMPLE_INTEGER := 1;

    PROCEDURE process_dept(in_dept_id IN departments.department_id%TYPE) IS
    BEGIN
        NULL;
    END process_dept;

    PROCEDURE remove_employee (in_employee_id IN employees.employee_id%TYPE) IS
        l_dept_id      employees.department_id%TYPE;
        l_deleted_emps SIMPLE_INTEGER;
    BEGIN
        DELETE FROM employees
            WHERE employee_id = in_employee_id
            RETURNING department_id INTO l_dept_id;

        l_deleted_emps := SQL%ROWCOUNT;

        process_dept(in_dept_id => l_dept_id);

        IF l_deleted_emps > co_one THEN
            -- too many rows deleted.
            ROLLBACK;
        END IF;
    END remove_employee;
END employee_api;
/

```

## CASE / IF / DECODE / NVL / NVL2 / COALESCE

**G-4210: Try to use CASE rather than an IF statement with multiple ELSIF paths.**

### Major

Maintainability, Testability

#### REASON

IF statements containing multiple ELSIF tend to become complex quickly.

#### EXAMPLE (BAD)

```
DECLARE
    l_color VARCHAR2(7 CHAR);
BEGIN
    IF l_color = constants_up.co_red THEN
        my_package.do_red();
    ELSIF l_color = constants_up.co_blue THEN
        my_package.do_blue();
    ELSIF l_color = constants_up.co_black THEN
        my_package.do_black();
    END IF;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    l_color types_up.color_code_type;
BEGIN
    CASE l_color
        WHEN constants_up.co_red THEN
            my_package.do_red();
        WHEN constants_up.co_blue THEN
            my_package.do_blue();
        WHEN constants_up.co_black THEN
            my_package.do_black();
        ELSE NULL;
    END CASE;
END;
/
```

## G-4220: Try to use CASE rather than DECODE.



Minor

Maintainability, Portability

### REASON

DECODE is an ORACLE specific function hard to understand and restricted to SQL only. The “newer” CASE function is much more common has a better readability and may be used within PL/SQL too.

### EXAMPLE (BAD)

```
SELECT DECODE(dummy, 'X', 1
                    , 'Y', 2
                    , 'Z', 3
                    , 0)
FROM dual;
```

### EXAMPLE (GOOD)

```
SELECT CASE dummy
        WHEN 'X' THEN 1
        WHEN 'Y' THEN 2
        WHEN 'Z' THEN 3
        ELSE 0
END
FROM dual;
```

**G-4230: Always use a COALESCE instead of a NVL command, if parameter 2 of the NVL function is a function call or a SELECT statement.**

 **Critical**

Efficiency, Reliability

**REASON**

The `NVL` function always evaluates both parameters before deciding which one to use. This can be harmful if parameter 2 is either a function call or a select statement, as it will be executed regardless of whether parameter 1 contains a NULL value or not.

The `COALESCE` function does not have this drawback.

**EXAMPLE (BAD)**

```
SELECT NVL(dummy, my_package.expensive_null(value_in => dummy))  
FROM dual;
```

**EXAMPLE (GOOD)**

```
SELECT COALESCE(dummy, my_package.expensive_null(value_in => dummy))  
FROM dual;
```

**G-4240: Always use a CASE instead of a NVL2 command if parameter 2 or 3 of NVL2 is either a function call or a SELECT statement.**

 **Critical**

Efficiency, Reliability

**REASON**

The `NVL2` function always evaluates all parameters before deciding which one to use. This can be harmful, if parameter 2 or 3 is either a function call or a select statement, as they will be executed regardless of whether parameter 1 contains a `NULL` value or not.

**EXAMPLE (BAD)**

```
SELECT NVL2(dummy, my_package.expensive_nn(value_in => dummy),
            my_package.expensive_null(value_in => dummy))
FROM dual;
```

**EXAMPLE (GOOD)**

```
SELECT CASE
    WHEN dummy IS NULL THEN
        my_package.expensive_null(value_in => dummy)
    ELSE
        my_package.expensive_nn(value_in => dummy)
END
FROM dual;
```

## Flow Control

### G-4310: Never use GOTO statements in your code.

#### Major

Maintainability, Testability

#### REASON

Code containing gotos is hard to format. Indentation should be used to show logical structure, and gotos have an effect on logical structure. Using indentation to show the logical structure of a goto and its target, however, is difficult or impossible. (...)

Use of gotos is a matter of religion. My dogma is that in modern languages, you can easily replace nine out of ten gotos with equivalent sequential constructs. In these simple cases, you should replace gotos out of habit. In the hard cases, you can still exorcise the goto in nine out of ten cases: You can break the code into smaller routines, use try-finally, use nested ifs, test and retest a status variable, or restructure a conditional. Eliminating the goto is harder in these cases, but it's good mental exercise (...).

– McConnell, Steve C. (2004). *Code Complete. Second Edition*. Microsoft Press.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE password_check (in_password IN VARCHAR2) IS
    co_digitarray  CONSTANT STRING(10 CHAR)  := '0123456789';
    co_lower_bound CONSTANT SIMPLE_INTEGER := 1;
    co_errno       CONSTANT SIMPLE_INTEGER := -20501;
    co_errmsg      CONSTANT STRING(100 CHAR) := 'Password must contain a digit.';
    l_isdigit      BOOLEAN      := FALSE;
    l_len_pw       PLS_INTEGER;
    l_len_array    PLS_INTEGER;
  BEGIN
    l_len_pw      := LENGTH(in_password);
    l_len_array   := LENGTH(co_digitarray);

    <<check_digit>>
    FOR i IN co_lower_bound .. l_len_array
    LOOP
      <<check_pw_char>>
      FOR j IN co_lower_bound .. l_len_pw
      LOOP
        IF SUBSTR(in_password, j, 1) = SUBSTR(co_digitarray, i, 1) THEN
          l_isdigit := TRUE;
          GOTO check_other_things;
        END IF;
      END LOOP check_pw_char;
    END LOOP check_digit;

    <<check_other_things>>
    NULL;

    IF NOT l_isdigit THEN
      raise_application_error(co_errno, co_errmsg);
    END IF;
  END password_check;
END my_package;
/
```

#### EXAMPLE (BETTER)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE password_check (in_password IN VARCHAR2) IS
    co_digitarray  CONSTANT STRING(10 CHAR)  := '0123456789';
    co_lower_bound CONSTANT SIMPLE_INTEGER := 1;
    co_errno       CONSTANT SIMPLE_INTEGER := -20501;
    co_errmsg      CONSTANT STRING(100 CHAR) := 'Password must contain a digit.';
    l_isdigit      BOOLEAN                  := FALSE;
    l_len_pw       PLS_INTEGER;
    l_len_array    PLS_INTEGER;
  BEGIN
    l_len_pw      := LENGTH(in_password);
    l_len_array   := LENGTH(co_digitarray);

    <<check_digit>>
    FOR i IN co_lower_bound .. l_len_array
    LOOP
      <<check_pw_char>>
      FOR j IN co_lower_bound .. l_len_pw
      LOOP
        IF SUBSTR(in_password, j, 1) = SUBSTR(co_digitarray, i, 1) THEN
          l_isdigit := TRUE;
          EXIT check_digit; -- early exit condition
        END IF;
      END LOOP check_pw_char;
    END LOOP check_digit;

    <<check_other_things>>
    NULL;

    IF NOT l_isdigit THEN
      raise_application_error(co_errno, co_errmsg);
    END IF;
  END password_check;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE password_check (in_password IN VARCHAR2) IS
    co_digitpattern CONSTANT STRING(10 CHAR) := '\d';
    co_errno        CONSTANT SIMPLE_INTEGER := -20501;
    co_errmsg       CONSTANT STRING(100 CHAR) := 'Password must contain a digit.';
  BEGIN
    IF NOT REGEXP_LIKE(in_password, co_digitpattern)
    THEN
      raise_application_error(co_errno, co_errmsg);
    END IF;
  END password_check;
END my_package;
/
```



## G-4320: Always label your loops.



Minor

Maintainability

### REASON

It's a good alternative for comments to indicate the start and end of a named loop processing.

### EXAMPLE (BAD)

```
DECLARE
  i INTEGER;
  co_min_value CONSTANT SIMPLE_INTEGER := 1;
  co_max_value CONSTANT SIMPLE_INTEGER := 10;
  co_increment CONSTANT SIMPLE_INTEGER := 1;
BEGIN
  i := co_min_value;
  WHILE (i <= co_max_value)
  LOOP
    i := i + co_increment;
  END LOOP;

  LOOP
    EXIT;
  END LOOP;

  FOR i IN co_min_value..co_max_value
  LOOP
    sys.dbms_output.put_line(i);
  END LOOP;

  FOR r_employee IN (SELECT last_name FROM employees)
  LOOP
    sys.dbms_output.put_line(r_employee.last_name);
  END LOOP;
END;
/
```

### EXAMPLE (GOOD)

```

DECLARE
  i INTEGER;
  co_min_value CONSTANT SIMPLE_INTEGER := 1;
  co_max_value CONSTANT SIMPLE_INTEGER := 10;
  co_increment CONSTANT SIMPLE_INTEGER := 1;
BEGIN
  i := co_min_value;
  <<while_loop>>
  WHILE (i <= co_max_value)
  LOOP
    i := i + co_increment;
  END LOOP while_loop;

  <<basic_loop>>
  LOOP
    EXIT basic_loop;
  END LOOP basic_loop;

  <<for_loop>>
  FOR i IN co_min_value..co_max_value
  LOOP
    sys.dbms_output.put_line(i);
  END LOOP for_loop;

  <<process_employees>>
  FOR r_employee IN (SELECT last_name
                     FROM employees)
  LOOP
    sys.dbms_output.put_line(r_employee.last_name);
  END LOOP process_employees;
END;
/

```

**G-4330: Always use a CURSOR FOR loop to process the complete cursor results unless you are using bulk operations.**



Minor

Maintainability

**REASON**

It is easier for the reader to see, that the complete data set is processed. Using SQL to define the data to be processed is easier to maintain and typically faster than using conditional processing within the loop.

Since an `EXIT` statement is similar to a `GOTO` statement, it should be avoided, whenever possible.

**EXAMPLE (BAD)**

```
DECLARE
  CURSOR c_employees IS
    SELECT employee_id, last_name
      FROM employees;
  r_employee c_employees%ROWTYPE;
BEGIN
  OPEN c_employees;

  <<read_employees>>
  LOOP
    FETCH c_employees INTO r_employee;
    EXIT read_employees WHEN c_employees%NOTFOUND;
    sys.dbms_output.put_line(r_employee.last_name);
  END LOOP read_employees;

  CLOSE c_employees;
END;
/
```

**EXAMPLE (GOOD)**

```
DECLARE
  CURSOR c_employees IS
    SELECT employee_id, last_name
      FROM employees;
BEGIN
  <<read_employees>>
  FOR r_employee IN c_employees
  LOOP
    sys.dbms_output.put_line(r_employee.last_name);
  END LOOP read_employees;
END;
/
```

## G-4340: Always use a NUMERIC FOR loop to process a dense array.



Minor

Maintainability

### REASON

It is easier for the reader to see, that the complete array is processed.

Since an `EXIT` statement is similar to a `GOTO` statement, it should be avoided, whenever possible.

### EXAMPLE (BAD)

```
DECLARE
  TYPE t_employee_type IS VARRAY(10) OF employees.employee_id%TYPE;
  t_employees t_employee_type;
  co_himuro    CONSTANT INTEGER := 118;
  co_livingston CONSTANT INTEGER := 177;
  co_min_value CONSTANT SIMPLE_INTEGER := 1;
  co_increment CONSTANT SIMPLE_INTEGER := 1;
  i PLS_INTEGER;
BEGIN
  t_employees := t_employee_type(co_himuro, co_livingston);
  i          := co_min_value;

  <<process_employees>>
  LOOP
    EXIT process_employees WHEN i > t_employees.COUNT();
    sys.dbms_output.put_line(t_employees(i));
    i := i + co_increment;
  END LOOP process_employees;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  TYPE t_employee_type IS VARRAY(10) OF employees.employee_id%TYPE;
  t_employees t_employee_type;
  co_himuro    CONSTANT INTEGER := 118;
  co_livingston CONSTANT INTEGER := 177;
BEGIN
  t_employees := t_employee_type(co_himuro, co_livingston);

  <<process_employees>>
  FOR i IN 1..t_employees.COUNT()
  LOOP
    sys.dbms_output.put_line(t_employees(i));
  END LOOP process_employees;
END;
/
```

**G-4350: Always use 1 as lower and COUNT() as upper bound when looping through a dense array.**

**⚠ Major**

Reliability

**REASON**

Doing so will not raise a `VALUE_ERROR` if the array you are looping through is empty. If you want to use `FIRST()..LAST()` you need to check the array for emptiness beforehand to avoid the raise of `VALUE_ERROR`.

**EXAMPLE (BAD)**

```
DECLARE
  TYPE t_employee_type IS TABLE OF employees.employee_id%TYPE;
  t_employees t_employee_type := t_employee_type();
BEGIN
  <<process_employees>>
  FOR i IN t_employees.FIRST()..t_employees.LAST()
  LOOP
    sys.dbms_output.put_line(t_employees(i)); -- some processing
  END LOOP process_employees;
END;
/
```

**EXAMPLE (BETTER)**

Raise an uninitialized collection error if `t_employees` is not initialized.

```
DECLARE
  TYPE t_employee_type IS TABLE OF employees.employee_id%TYPE;
  t_employees t_employee_type := t_employee_type();
BEGIN
  <<process_employees>>
  FOR i IN 1..t_employees.COUNT()
  LOOP
    sys.dbms_output.put_line(t_employees(i)); -- some processing
  END LOOP process_employees;
END;
/
```

**EXAMPLE (GOOD)**

Raises neither an error nor checking whether the array is empty. `t_employees.COUNT()` always returns a `NUMBER` (unless the array is not initialized). If the array is empty `COUNT()` returns 0 and therefore the loop will not be entered.

```
DECLARE
  TYPE t_employee_type IS TABLE OF employees.employee_id%TYPE;
  t_employees t_employee_type := t_employee_type();
BEGIN
  IF t_employees IS NOT NULL THEN
    <<process_employees>>
    FOR i IN 1..t_employees.COUNT()
    LOOP
      sys.dbms_output.put_line(t_employees(i)); -- some processing
    END LOOP process_employees;
  END IF;
END;
/
```

## G-4360: Always use a WHILE loop to process a loose array.



Minor

Efficiency

### REASON

When a loose array is processed using a `NUMERIC FOR LOOP` we have to check with all iterations whether the element exist to avoid a `NO_DATA_FOUND` exception. In addition, the number of iterations is not driven by the number of elements in the array but by the number of the lowest/highest element. The more gaps we have, the more superfluous iterations will be done.

### EXAMPLE (BAD)

```
DECLARE -- raises no_data_found when processing 2nd record
  TYPE t_employee_type IS TABLE OF employees.employee_id%TYPE;
  t_employees t_employee_type;
  co_rogers    CONSTANT INTEGER := 134;
  co_matos     CONSTANT INTEGER := 143;
  co_mcewen    CONSTANT INTEGER := 158;
  co_index_matos CONSTANT INTEGER := 2;
BEGIN
  t_employees := t_employee_type(co_rogers, co_matos, co_mcewen);
  t_employees.DELETE(co_index_matos);

  IF t_employees IS NOT NULL THEN
    <<process_employees>>
    FOR i IN 1..t_employees.COUNT()
      LOOP
        sys.dbms_output.put_line(t_employees(i));
      END LOOP process_employees;
  END IF;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  TYPE t_employee_type IS TABLE OF employees.employee_id%TYPE;
  t_employees t_employee_type;
  co_rogers    CONSTANT INTEGER := 134;
  co_matos     CONSTANT INTEGER := 143;
  co_mcewen    CONSTANT INTEGER := 158;
  co_index_matos CONSTANT INTEGER := 2;
  l_index      PLS_INTEGER;
BEGIN
  t_employees := t_employee_type(co_rogers, co_matos, co_mcewen);
  t_employees.DELETE(co_index_matos);

  l_index := t_employees.FIRST();

  <<process_employees>>
  WHILE l_index IS NOT NULL
    LOOP
      sys.dbms_output.put_line(t_employees(l_index));
      l_index := t_employees.NEXT(l_index);
    END LOOP process_employees;
END;
/
```

## G-4370: Avoid using EXIT to stop loop processing unless you are in a basic loop.

### Major

Maintainability

#### REASON

A numeric for loop as well as a while loop and a cursor for loop have defined loop boundaries. If you are not able to exit your loop using those loop boundaries, then a basic loop is the right loop to choose.

#### EXAMPLE (BAD)

```
DECLARE
  i INTEGER;
  co_min_value CONSTANT SIMPLE_INTEGER := 1;
  co_max_value CONSTANT SIMPLE_INTEGER := 10;
  co_increment CONSTANT SIMPLE_INTEGER := 1;
BEGIN
  i := co_min_value;
  <<while_loop>>
  WHILE (i <= co_max_value)
  LOOP
    i := i + co_increment;
    EXIT while_loop WHEN i > co_max_value;
  END LOOP while_loop;

  <<basic_loop>>
  LOOP
    EXIT basic_loop;
  END LOOP basic_loop;

  <<for_loop>>
  FOR i IN co_min_value..co_max_value
  LOOP
    NULL;
    EXIT for_loop WHEN i = co_max_value;
  END LOOP for_loop;

  <<process_employees>>
  FOR r_employee IN (SELECT last_name
                     FROM employees)
  LOOP
    sys.dbms_output.put_line(r_employee.last_name);
    NULL; -- some processing
    EXIT process_employees;
  END LOOP process_employees;
END;
/
```

#### EXAMPLE (GOOD)

```

DECLARE
    i INTEGER;
    co_min_value CONSTANT SIMPLE_INTEGER := 1;
    co_max_value CONSTANT SIMPLE_INTEGER := 10;
    co_increment CONSTANT SIMPLE_INTEGER := 1;
BEGIN
    i := co_min_value;
    <<while_loop>>
    WHILE (i <= co_max_value)
    LOOP
        i := i + co_increment;
    END LOOP while_loop;

    <<basic_loop>>
    LOOP
        EXIT basic_loop;
    END LOOP basic_loop;

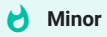
    <<for_loop>>
    FOR i IN co_min_value..co_max_value
    LOOP
        sys.dbms_output.put_line(i);
    END LOOP for_loop;

    <<process_employees>>
    FOR r_employee IN (SELECT last_name
                        FROM employees)
    LOOP
        sys.dbms_output.put_line(r_employee.last_name); -- some processing
    END LOOP process_employees;
END;
/

```



## G-4375: Always use EXIT WHEN instead of an IF statement to exit from a loop.



Minor

Maintainability

### REASON

If you need to use an `EXIT` statement use its full semantic to make the code easier to understand and maintain. There is simply no need for an additional IF statement.

### EXAMPLE (BAD)

```
DECLARE
    co_first_year CONSTANT PLS_INTEGER := 1900;
BEGIN
    <<process_employees>>
    LOOP
        my_package.some_processing();

        IF EXTRACT(year FROM SYSDATE) > co_first_year THEN
            EXIT process_employees;
        END IF;

        my_package.some_further_processing();
    END LOOP process_employees;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
    co_first_year CONSTANT PLS_INTEGER := 1900;
BEGIN
    <<process_employees>>
    LOOP
        my_package.some_processing();

        EXIT process_employees WHEN EXTRACT(YEAR FROM SYSDATE) > co_first_year;

        my_package.some_further_processing();
    END LOOP process_employees;
END;
/
```

## G-4380 Try to label your EXIT WHEN statements.



Minor

Maintainability

### REASON

It's a good alternative for comments, especially for nested loops to name the loop to exit.

### EXAMPLE (BAD)

```
DECLARE
  co_init_loop  CONSTANT SIMPLE_INTEGER      := 0;
  co_increment  CONSTANT SIMPLE_INTEGER      := 1;
  co_exit_value CONSTANT SIMPLE_INTEGER      := 3;
  co_outer_text CONSTANT types_up.short_text_type := 'Outer Loop counter is ';
  co_inner_text CONSTANT types_up.short_text_type := ' Inner Loop counter is ';
  l_outerlp    PLS_INTEGER;
  l_innerlp    PLS_INTEGER;
BEGIN
  l_outerlp := co_init_loop;
  <<outerloop>>
  LOOP
    l_innerlp := co_init_loop;
    l_outerlp := NVL(l_outerlp, co_init_loop) + co_increment;
    <<innerloop>>
    LOOP
      l_innerlp := NVL(l_innerlp, co_init_loop) + co_increment;
      sys.dbms_output.put_line(co_outer_text || l_outerlp ||
                               co_inner_text || l_innerlp);

      EXIT WHEN l_innerlp = co_exit_value;
    END LOOP innerloop;

    EXIT WHEN l_innerlp = co_exit_value;
  END LOOP outerloop;
END;
/
```

### EXAMPLE (GOOD)

```

DECLARE
  co_init_loop  CONSTANT SIMPLE_INTEGER      := 0;
  co_increment  CONSTANT SIMPLE_INTEGER      := 1;
  co_exit_value CONSTANT SIMPLE_INTEGER      := 3;
  co_outer_text CONSTANT types_up.short_text_type := 'Outer Loop counter is ';
  co_inner_text CONSTANT types_up.short_text_type := ' Inner Loop counter is ';
  l_outerlp    PLS_INTEGER;
  l_innerlp    PLS_INTEGER;
BEGIN
  l_outerlp := co_init_loop;
  <<outerloop>>
  LOOP
    l_innerlp := co_init_loop;
    l_outerlp := NVL(l_outerlp,co_init_loop) + co_increment;
    <<innerloop>>
    LOOP
      l_innerlp := NVL(l_innerlp, co_init_loop) + co_increment;
      sys.dbms_output.put_line(co_outer_text || l_outerlp ||
                              co_inner_text || l_innerlp);

      EXIT outerloop WHEN l_innerlp = co_exit_value;
    END LOOP innerloop;
  END LOOP outerloop;
END;
/

```

**G-4385: Never use a cursor for loop to check whether a cursor returns data.**

**⚠ Major**

Efficiency

**REASON**

You might process more data than required, which leads to bad performance.

**EXAMPLE (BAD)**

```
DECLARE
  l_employee_found BOOLEAN := FALSE;
  CURSOR c_employees IS
    SELECT employee_id, last_name
      FROM employees;
BEGIN
  <<check_employees>>
  FOR r_employee IN c_employees
  LOOP
    l_employee_found := TRUE;
  END LOOP check_employees;
END;
/
```

**EXAMPLE (GOOD)**

```
DECLARE
  l_employee_found BOOLEAN := FALSE;
  CURSOR c_employees IS
    SELECT employee_id, last_name
      FROM employees;
  r_employee c_employees%ROWTYPE;
BEGIN
  OPEN c_employees;
  FETCH c_employees INTO r_employee;
  l_employee_found := c_employees%FOUND;
  CLOSE c_employees;
END;
/
```

## G-4390: Avoid use of unreferenced FOR loop indexes.

### Major

Efficiency

#### REASON

If the loop index is used for anything but traffic control inside the loop, this is one of the indicators that a numeric FOR loop is being used incorrectly. The actual body of executable statements completely ignores the loop index. When that is the case, there is a good chance that you do not need the loop at all.

#### EXAMPLE (BAD)

```
DECLARE
  l_row    PLS_INTEGER;
  l_value  PLS_INTEGER;
  co_lower_bound CONSTANT SIMPLE_INTEGER      := 1;
  co_upper_bound CONSTANT SIMPLE_INTEGER      := 5;
  co_row_incr  CONSTANT SIMPLE_INTEGER      := 1;
  co_value_incr CONSTANT SIMPLE_INTEGER      := 10;
  co_delimiter  CONSTANT types_up.short_text_type := ' ';
  co_first_value CONSTANT SIMPLE_INTEGER      := 100;
BEGIN
  l_row := co_lower_bound;
  l_value := co_first_value;
  <<for_loop>>
  FOR i IN co_lower_bound .. co_upper_bound
  LOOP
    sys.dbms_output.put_line(l_row || co_delimiter || l_value);
    l_row := l_row + co_row_incr;
    l_value := l_value + co_value_incr;
  END LOOP for_loop;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
  co_lower_bound CONSTANT SIMPLE_INTEGER      := 1;
  co_upper_bound CONSTANT SIMPLE_INTEGER      := 5;
  co_value_incr  CONSTANT SIMPLE_INTEGER      := 10;
  co_delimiter  CONSTANT types_up.short_text_type := ' ';
  co_first_value CONSTANT SIMPLE_INTEGER      := 100;
BEGIN
  <<for_loop>>
  FOR i IN co_lower_bound .. co_upper_bound
  LOOP
    sys.dbms_output.put_line(i || co_delimiter ||
                             to_char(co_first_value + i * co_value_incr));
  END LOOP for_loop;
END;
/
```

## G-4395: Avoid hard-coded upper or lower bound values with FOR loops.



Minor

Changeability, Maintainability

### REASON

Your `LOOP` statement uses a hard-coded value for either its upper or lower bounds. This creates a "weak link" in your program because it assumes that this value will never change. A better practice is to create a named constant (or function) and reference this named element instead of the hard-coded value.

### EXAMPLE (BAD)

```
BEGIN
  <<for_loop>>
  FOR i IN 1..5
  LOOP
    sys.dbms_output.put_line(i);
  END LOOP for_loop;
END;
/
```

### EXAMPLE (GOOD)

```
DECLARE
  co_lower_bound CONSTANT SIMPLE_INTEGER := 1;
  co_upper_bound CONSTANT SIMPLE_INTEGER := 5;
BEGIN
  <<for_loop>>
  FOR i IN co_lower_bound..co_upper_bound
  LOOP
    sys.dbms_output.put_line(i);
  END LOOP for_loop;
END;
/
```

# Exception Handling

G-5010: Try to use a error/logging framework for your application.

## Critical

Reliability, Reusability, Testability

### Reason

Having a framework to raise/handle/log your errors allows you to easily avoid duplicate application error numbers and having different error messages for the same type of error.

This kind of framework should include

- Logging (different channels like table, mail, file, etc. if needed)
- Error Raising
- Multilanguage support if needed
- Translate ORACLE error messages to a user friendly error text
- Error repository

### Example (bad)

```
BEGIN
  sys.dbms_output.put_line('START');
  -- some processing
  sys.dbms_output.put_line('END');
END;
/
```

### Example (good)

```
DECLARE
  -- see https://github.com/OraOpenSource/Logger
  l_scope logger_logs.scope%type := 'DEMO';
BEGIN
  logger.log('START', l_scope);
  -- some processing
  logger.log('END', l_scope);
END;
/
```

## G-5020: Never handle unnamed exceptions using the error number.

### Critical

Maintainability

#### Reason

When literals are used for error numbers the reader needs the error message manual to understand what is going on. Commenting the code or using constants is an option, but it is better to use named exceptions instead, because it ensures a certain level of consistency which makes maintenance easier.

#### Example (bad)

```
DECLARE
    co_no_data_found CONSTANT INTEGER := -1;
BEGIN
    my_package.some_processing(); -- some code which raises an exception
EXCEPTION
    WHEN TOO_MANY_ROWS THEN
        my_package.some_further_processing();
    WHEN OTHERS THEN
        IF SQLCODE = co_no_data_found THEN
            NULL;
        END IF;
END;
/
```

#### Example (good)

```
BEGIN
    my_package.some_processing(); -- some code which raises an exception
EXCEPTION
    WHEN TOO_MANY_ROWS THEN
        my_package.some_further_processing();
    WHEN NO_DATA_FOUND THEN
        NULL; -- handle no_data_found
END;
/
```



## G-5030: Never assign predefined exception names to user defined exceptions.

### Blocker

Reliability, Testability

#### Reason

This is error-prone because your local declaration overrides the global declaration. While it is technically possible to use the same names, it causes confusion for others needing to read and maintain this code. Additionally, you will need to be very careful to use the prefix `STANDARD` in front of any reference that needs to use Oracle's default exception behavior.

#### Example (bad)

Using the code below, we are not able to handle the `no_data_found` exception raised by the `SELECT` statement as we have overwritten that exception handler. In addition, our exception handler doesn't have an exception number assigned, which should be raised when the `SELECT` statement does not find any rows.

```
DECLARE
  l_dummy dual.dummy%TYPE;
  no_data_found EXCEPTION;
  co_rownum CONSTANT SIMPLE_INTEGER := 0;
  co_no_data_found CONSTANT types_up.short_text_type := 'no_data_found';
BEGIN
  SELECT dummy
    INTO l_dummy
   FROM dual
  WHERE ROWNUM = co_rownum;

  IF l_dummy IS NULL THEN
    RAISE no_data_found;
  END IF;
EXCEPTION
  WHEN no_data_found THEN
    sys.dbms_output.put_line(co_no_data_found);
END;
/

Error report -
ORA-01403: no data found
ORA-06512: at line 5
01403. 00000 - "no data found"
*Cause:      No data was found from the objects.
*Action:     There was no data from the objects which may be due to end of fetch.
```

#### Example (good)

```

DECLARE
    l_dummy dual.dummy%TYPE;
    empty_value      EXCEPTION;
    co_rownum        CONSTANT simple_integer      := 0;
    co_empty_value    CONSTANT types_up.short_text_type := 'empty_value';
    co_no_data_found  CONSTANT types_up.short_text_type := 'no_data_found';
BEGIN
    SELECT dummy
       INTO l_dummy
    FROM dual
    WHERE rownum = co_rownum;

    IF l_dummy IS NULL THEN
        RAISE empty_value;
    END IF;
EXCEPTION
    WHEN empty_value THEN
        sys.dbms_output.put_line(co_empty_value);
    WHEN no_data_found THEN
        sys.dbms_output.put_line(co_no_data_found);
END;
/

```

## G-5040: Avoid use of WHEN OTHERS clause in an exception section without any other specific handlers.

### Major

Reliability

#### Reason

There is not necessarily anything wrong with using `WHEN OTHERS`, but it can cause you to "lose" error information unless your handler code is relatively sophisticated. Generally, you should use `WHEN OTHERS` to grab any and every error only after you have thought about your executable section and decided that you are not able to trap any specific exceptions. If you know, on the other hand, that a certain exception might be raised, include a handler for that error. By declaring two different exception handlers, the code more clearly states what we expect to have happen and how we want to handle the errors. That makes it easier to maintain and enhance. We also avoid hard-coding error numbers in checks against `SQLCODE`.

#### Example (bad)

```
BEGIN
    my_package.some_processing();
EXCEPTION
    WHEN OTHERS THEN
        my_package.some_further_processing();
END;
/
```

#### Example (good)

```
BEGIN
    my_package.some_processing();
EXCEPTION
    WHEN DUP_VAL_ON_INDEX THEN
        my_package.some_further_processing();
END;
/
```

## G-5050: Avoid use of the RAISE\_APPLICATION\_ERROR built-in procedure with a hard-coded 20nnn error number or hard-coded message.

### Major

Changeability, Maintainability

#### Reason

If you are not very organized in the way you allocate, define and use the error numbers between 20999 and 20000 (those reserved by Oracle for its user community), it is very easy to end up with conflicting usages. You should assign these error numbers to named constants and consolidate all definitions within a single package. When you call

`RAISE_APPLICATION_ERROR`, you should reference these named elements and error message text stored in a table. Use your own raise procedure in place of explicit calls to `RAISE_APPLICATION_ERROR`. If you are raising a "system" exception like `NO_DATA_FOUND`, you must use `RAISE`. However, when you want to raise an application-specific error, you use `RAISE_APPLICATION_ERROR`. If you use the latter, you then have to provide an error number and message. This leads to unnecessary and damaging hard-coded values. A more fail-safe approach is to provide a predefined raise procedure that automatically checks the error number and determines the correct way to raise the error.

#### Example (bad)

```
BEGIN
    raise_application_error(-20501,'Invalid employee_id');
END;
/
```

#### Example (good)

```
BEGIN
    err_up.raise(in_error => err.co_invalid_employee_id);
END;
/
```

## G-5060: Avoid unhandled exceptions.

### Major

Reliability

#### Reason

This may be your intention, but you should review the code to confirm this behavior.

If you are raising an error in a program, then you are clearly predicting a situation in which that error will occur. You should consider including a handler in your code for predictable errors, allowing for a graceful and informative failure. After all, it is much more difficult for an enclosing block to be aware of the various errors you might raise and more importantly, what should be done in response to the error.

The form that this failure takes does not necessarily need to be an exception. When writing functions, you may well decide that in the case of certain exceptions, you will want to return a value such as NULL, rather than allow an exception to propagate out of the function.

#### Example (bad)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  FUNCTION name_by_id (in_id IN departments.department_id%TYPE)
    RETURN departments.department_name%TYPE IS
    l_department_name departments.department_name%TYPE;
  BEGIN
    SELECT department_name
      INTO l_department_name
      FROM departments
      WHERE department_id = in_id;

    RETURN l_department_name;
  END name_by_id;
END department_api;
/
```

#### Example (good)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  FUNCTION name_by_id (in_id IN departments.department_id%TYPE)
    RETURN departments.department_name%TYPE IS
    l_department_name departments.department_name%TYPE;
  BEGIN
    SELECT department_name
      INTO l_department_name
      FROM departments
      WHERE department_id = in_id;

    RETURN l_department_name;
  EXCEPTION
    WHEN NO_DATA_FOUND THEN RETURN NULL;
    WHEN TOO_MANY_ROWS THEN RAISE;
  END name_by_id;
END department_api;
/
```

## G-5070: Avoid using Oracle predefined exceptions.

 **Critical**

Reliability

### Reason

You have raised an exception whose name was defined by Oracle. While it is possible that you have a good reason for "using" one of Oracle's predefined exceptions, you should make sure that you would not be better off declaring your own exception and raising that instead.

If you decide to change the exception you are using, you should apply the same consideration to your own exceptions. Specifically, do not "re-use" exceptions. You should define a separate exception for each error condition, rather than use the same exception for different circumstances.

Being as specific as possible with the errors raised will allow developers to check for, and handle, the different kinds of errors the code might produce.

### Example (bad)

```
BEGIN
    RAISE NO_DATA_FOUND;
END;
/
```

### Example (good)

```
DECLARE
    my_exception EXCEPTION;
BEGIN
    RAISE my_exception;
END;
/
```

## Dynamic SQL

G-6010: Always use a character variable to execute dynamic SQL.

### Major

Maintainability, Testability

#### Reason

Having the executed statement in a variable makes it easier to debug your code (e.g. by logging the statement that failed).

#### Example (bad)

```
DECLARE
    l_next_val employees.employee_id%TYPE;
BEGIN
    EXECUTE IMMEDIATE 'select employees_seq.nextval from dual' INTO l_next_val;
END;
/
```

#### Example (good)

```
DECLARE
    l_next_val employees.employee_id%TYPE;
    co_sql CONSTANT types_up.big_string_type :=
        'select employees_seq.nextval from dual';
BEGIN
    EXECUTE IMMEDIATE co_sql INTO l_next_val;
END;
/
```

G-6020: Try to use output bind arguments in the RETURNING INTO clause of dynamic DML statements rather than the USING clause.

#### Minor

Maintainability

### Reason

When a dynamic INSERT , UPDATE , or DELETE statement has a RETURNING clause, output bind arguments can go in the RETURNING INTO clause or in the USING clause.

You should use the RETURNING INTO clause for values returned from a DML operation. Reserve OUT and IN OUT bind variables for dynamic PL/SQL blocks that return values in PL/SQL variables.

### Example (bad)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
    PROCEDURE upd_salary (in_employee_id IN      employees.employee_id%TYPE
                          ,in_increase_pct IN    types_up.percentage
                          ,out_new_salary      OUT employees.salary%TYPE)
    IS
        co_sql_stmt CONSTANT types_up.big_string_type := '
            UPDATE employees SET salary = salary + (salary / 100 * :1)
            WHERE employee_id = :2
            RETURNING salary INTO :3';
    BEGIN
        EXECUTE IMMEDIATE co_sql_stmt
            USING in_increase_pct, in_employee_id, OUT out_new_salary;
    END upd_salary;
END employee_api;
/
```

### Example (good)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
    PROCEDURE upd_salary (in_employee_id IN      employees.employee_id%TYPE
                          ,in_increase_pct IN    types_up.percentage
                          ,out_new_salary      OUT employees.salary%TYPE)
    IS
        co_sql_stmt CONSTANT types_up.big_string_type :=
            'UPDATE employees SET salary = salary + (salary / 100 * :1)
            WHERE employee_id = :2
            RETURNING salary INTO :3';
    BEGIN
        EXECUTE IMMEDIATE co_sql_stmt
            USING in_increase_pct, in_employee_id
            RETURNING INTO out_new_salary;
    END upd_salary;
END employee_api;
/
```



# Stored Objects

## General

**G-7110: Try to use named notation when calling program units.**

### Major

Changeability, Maintainability

#### REASON

Named notation makes sure that changes to the signature of the called program unit do not affect your call.

This is not needed for standard functions like (`TO_CHAR`, `TO_DATE`, `NVL`, `ROUND`, etc.) but should be followed for any other stored object having more than one parameter.

#### EXAMPLE (BAD)

```
DECLARE
    r_employee employees%rowtype;
    co_id CONSTANT employees.employee_id%type := 107;
BEGIN
    employee_api.employee_by_id(r_employee, co_id);
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    r_employee employees%rowtype;
    co_id CONSTANT employees.employee_id%type := 107;
BEGIN
    employee_api.employee_by_id(out_row => r_employee, in_employee_id => co_id);
END;
/
```

## G-7120 Always add the name of the program unit to its end keyword.



Minor

Maintainability

### REASON

It's a good alternative for comments to indicate the end of program units, especially if they are lengthy or nested.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
  FUNCTION employee_by_id (in_employee_id IN employees.employee_id%TYPE)
    RETURN employees%rowtype IS
    r_employee employees%rowtype;
  BEGIN
    SELECT *
      INTO r_employee
      FROM employees
      WHERE employee_id = in_employee_id;

    RETURN r_employee;
  EXCEPTION
    WHEN NO_DATA_FOUND THEN
      NULL;
    WHEN TOO_MANY_ROWS THEN
      RAISE;
  END;
END;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
  FUNCTION employee_by_id (in_employee_id IN employees.employee_id%TYPE)
    RETURN employees%rowtype IS
    r_employee employees%rowtype;
  BEGIN
    SELECT *
      INTO r_employee
      FROM employees
      WHERE employee_id = in_employee_id;

    RETURN r_employee;
  EXCEPTION
    WHEN NO_DATA_FOUND THEN
      NULL;
    WHEN TOO_MANY_ROWS THEN
      RAISE;
  END employee_by_id;
END employee_api;
/
```

## G-7130: Always use parameters or pull in definitions rather than referencing external variables in a local program unit.

### Major

Maintainability, Reliability, Testability

#### REASON

Local procedures and functions offer an excellent way to avoid code redundancy and make your code more readable (and thus more maintainable). Your local program refers, however, an external data structure, i.e., a variable that is declared outside of the local program. Thus, it is acting as a global variable inside the program.

This external dependency is hidden, and may cause problems in the future. You should instead add a parameter to the parameter list of this program and pass the value through the list. This technique makes your program more reusable and avoids scoping problems, i.e. the program unit is less tied to particular variables in the program. In addition, unit encapsulation makes maintenance a lot easier and cheaper.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY EMPLOYEE_API IS
  PROCEDURE calc_salary (in_employee_id IN employees.employee_id%TYPE) IS
    r_emp employees%rowtype;

  FUNCTION commission RETURN NUMBER IS
    l_commission employees.salary%TYPE := 0;
  BEGIN
    IF r_emp.commission_pct IS NOT NULL
    THEN
      l_commission := r_emp.salary * r_emp.commission_pct;
    END IF;

    RETURN l_commission;
  END commission;
  BEGIN
    SELECT *
      INTO r_emp
      FROM employees
     WHERE employee_id = in_employee_id;

    SYS.DBMS_OUTPUT.PUT_LINE(r_emp.salary + commission());
  EXCEPTION
    WHEN NO_DATA_FOUND THEN
      NULL;
    WHEN TOO_MANY_ROWS THEN
      NULL;
  END calc_salary;
END employee_api;
/
```

#### EXAMPLE (GOOD)

```

CREATE OR REPLACE PACKAGE BODY EMPLOYEE_API IS
  PROCEDURE calc_salary (in_employee_id IN employees.employee_id%TYPE) IS
    r_emp employees%rowtype;

    FUNCTION commission (in_salary    IN employees.salary%TYPE
                        ,in_comm_pct IN employees.commission_pct%TYPE)
      RETURN NUMBER IS
        l_commission employees.salary%TYPE := 0;
    BEGIN
      IF in_comm_pct IS NOT NULL THEN
        l_commission := in_salary * in_comm_pct;
      END IF;

      RETURN l_commission;
    END commission;
  BEGIN
    SELECT *
      INTO r_emp
      FROM employees
     WHERE employee_id = in_employee_id;

    SYS.DBMS_OUTPUT.PUT_LINE(
      r_emp.salary + commission(in_salary => r_emp.salary
                              ,in_comm_pct => r_emp.commission_pct)
    );
  EXCEPTION
    WHEN NO_DATA_FOUND THEN
      NULL;
    WHEN TOO_MANY_ROWS THEN
      NULL;
  END calc_salary;
END employee_api;
/

```

## G-7140: Always ensure that locally defined procedures or functions are referenced.

### Major

Maintainability, Reliability

#### REASON

This can occur as the result of changes to code over time, but you should make sure that this situation does not reflect a problem. And you should remove the declaration to avoid maintenance errors in the future.

You should go through your programs and remove any part of your code that is no longer used. This is a relatively straightforward process for variables and named constants. Simply execute searches for a variable's name in that variable's scope. If you find that the only place it appears is in its declaration, delete the declaration.

There is never a better time to review all the steps you took, and to understand the reasons you took them, then immediately upon completion of your program. If you wait, you will find it particularly difficult to remember those parts of the program that were needed at one point, but were rendered unnecessary in the end.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_procedure IS
    FUNCTION my_func RETURN NUMBER IS
      co_true CONSTANT INTEGER := 1;
    BEGIN
      RETURN co_true;
    END my_func;
  BEGIN
    NULL;
  END my_procedure;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_procedure IS
    FUNCTION my_func RETURN NUMBER IS
      co_true CONSTANT INTEGER := 1;
    BEGIN
      RETURN co_true;
    END my_func;
  BEGIN
    sys.dbms_output.put_line(my_func());
  END my_procedure;
END my_package;
/
```

## G-7150: Try to remove unused parameters.



Minor

Efficiency, Maintainability

### REASON

You should go through your programs and remove any parameter that is no longer used.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
    FUNCTION name_by_id (in_department_id IN departments.department_id%TYPE
                        ,in_manager_id    IN departments.manager_id%TYPE)
        RETURN departments.department_name%TYPE IS
        l_department_name departments.department_name%TYPE;
    BEGIN
        <<find_department>>
        BEGIN
            SELECT department_name
            INTO l_department_name
            FROM departments
            WHERE department_id = in_department_id;
        EXCEPTION
            WHEN NO_DATA_FOUND OR TOO_MANY_ROWS THEN
                l_department_name := NULL;
        END find_department;

        RETURN l_department_name;
    END name_by_id;
END department_api;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
    FUNCTION name_by_id (in_department_id IN departments.department_id%TYPE)
        RETURN departments.department_name%TYPE IS
        l_department_name departments.department_name%TYPE;
    BEGIN
        <<find_department>>
        BEGIN
            SELECT department_name
            INTO l_department_name
            FROM departments
            WHERE department_id = in_department_id;
        EXCEPTION
            WHEN NO_DATA_FOUND OR TOO_MANY_ROWS THEN
                l_department_name := NULL;
        END find_department;

        RETURN l_department_name;
    END name_by_id;
END department_api;
/
```

# Packages

**G-7210: Try to keep your packages small. Include only few procedures and functions that are used in the same context.**

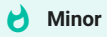
 **Minor**

Efficiency, Maintainability

**REASON**

The entire package is loaded into memory when the package is called the first time. To optimize memory consumption and keep load time small packages should be kept small but include components that are used together.

## G-7220: Always use forward declaration for private functions and procedures.



Minor

Changeability

### REASON

Having forward declarations allows you to order the functions and procedures of the package in a reasonable way.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE department_api IS
    PROCEDURE del (in_department_id IN departments.department_id%TYPE);
END department_api;
/

CREATE OR REPLACE PACKAGE BODY department_api IS
    FUNCTION does_exist (in_department_id IN departments.department_id%TYPE)
        RETURN BOOLEAN IS
        l_return PLS_INTEGER;
    BEGIN
        <<check_row_exists>>
        BEGIN
            SELECT 1
            INTO l_return
            FROM departments
            WHERE department_id = in_department_id;
        EXCEPTION
            WHEN no_data_found OR too_many_rows THEN
                l_return := 0;
        END check_row_exists;

        RETURN l_return = 1;
    END does_exist;

    PROCEDURE del (in_department_id IN departments.department_id%TYPE) IS
    BEGIN
        IF does_exist(in_department_id) THEN
            NULL;
        END IF;
    END del;
END department_api;
/
```

### EXAMPLE (GOOD)



```

CREATE OR REPLACE PACKAGE department_api IS
    PROCEDURE del (in_department_id IN departments.department_id%TYPE);
END department_api;
/

CREATE OR REPLACE PACKAGE BODY department_api IS
    FUNCTION does_exist (in_department_id IN departments.department_id%TYPE)
        RETURN BOOLEAN;

    PROCEDURE del (in_department_id IN departments.department_id%TYPE) IS
    BEGIN
        IF does_exist(in_department_id) THEN
            NULL;
        END IF;
    END del;

    FUNCTION does_exist (in_department_id IN departments.department_id%TYPE)
        RETURN BOOLEAN IS
        l_return PLS_INTEGER;
    BEGIN
        <<check_row_exists>>
        BEGIN
            SELECT 1
            INTO l_return
            FROM departments
            WHERE department_id = in_department_id;
        EXCEPTION
            WHEN no_data_found OR too_many_rows THEN
                l_return := 0;
        END check_row_exists;

        RETURN l_return = 1;
    END does_exist;
END department_api;
/

```

## G-7230: Avoid declaring global variables public.

### Major

Reliability

#### REASON

You should always declare package-level data inside the package body. You can then define "get and set" methods (functions and procedures, respectively) in the package specification to provide controlled access to that data. By doing so you can guarantee data integrity, you can change your data structure implementation, and also track access to those data structures.

Data structures (scalar variables, collections, cursors) declared in the package specification (not within any specific program) can be referenced directly by any program running in a session with EXECUTE rights to the package.

Instead, declare all package-level data in the package body and provide "get and set" methods - a function to get the value and a procedure to set the value - in the package specification. Developers then can access the data using these methods - and will automatically follow all rules you set upon data modification.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE employee_api AS
    co_min_increase CONSTANT types_up.sal_increase_type := 0.01;
    co_max_increase CONSTANT types_up.sal_increase_type := 0.5;
    g_salary_increase types_up.sal_increase_type := co_min_increase;

    PROCEDURE set_salary_increase (in_increase IN types_up.sal_increase_type);
    FUNCTION salary_increase RETURN types_up.sal_increase_type;
END employee_api;
/

CREATE OR REPLACE PACKAGE BODY employee_api AS
    PROCEDURE set_salary_increase (in_increase IN types_up.sal_increase_type) IS
    BEGIN
        g_salary_increase := GREATEST(LEAST(in_increase,co_max_increase)
                                     ,co_min_increase);
    END set_salary_increase;

    FUNCTION salary_increase RETURN types_up.sal_increase_type IS
    BEGIN
        RETURN g_salary_increase;
    END salary_increase;
END employee_api;
/
```

#### EXAMPLE (GOOD)

```

CREATE OR REPLACE PACKAGE employee_api AS
    PROCEDURE set_salary_increase (in_increase IN types_up.sal_increase_type);
    FUNCTION salary_increase RETURN types_up.sal_increase_type;
END employee_api;
/

CREATE OR REPLACE PACKAGE BODY employee_api AS
    g_salary_increase types_up.sal_increase_type(4,2);

    PROCEDURE init;

    PROCEDURE set_salary_increase (in_increase IN types_up.sal_increase_type) IS
    BEGIN
        g_salary_increase := GREATEST(LEAST(in_increase
                                           , constants_up.max_salary_increase())
                                       , constants_up.min_salary_increase());
    END set_salary_increase;

    FUNCTION salary_increase RETURN types_up.sal_increase_type IS
    BEGIN
        RETURN g_salary_increase;
    END salary_increase;

    PROCEDURE init
    IS
    BEGIN
        g_salary_increase := constants_up.min_salary_increase();
    END init;
BEGIN
    init();
END employee_api;
/

```

## G-7240: Avoid using an IN OUT parameter as IN or OUT only.

### Major

Efficiency, Maintainability

#### REASON

By showing the mode of parameters, you help the reader. If you do not specify a parameter mode, the default mode is `IN`. Explicitly showing the mode indication of all parameters is a more assertive action than simply taking the default mode. Anyone reviewing the code later will be more confident that you intended the parameter mode to be `IN` / `OUT`.

#### EXAMPLE (BAD)

```
-- Bad
CREATE OR REPLACE PACKAGE BODY employee_up IS
  PROCEDURE rcv_emp (io_first_name      IN OUT employees.first_name%TYPE
                    ,io_last_name       IN OUT employees.last_name%TYPE
                    ,io_email            IN OUT employees.email%TYPE
                    ,io_phone_number     IN OUT employees.phone_number%TYPE
                    ,io_hire_date        IN OUT employees.hire_date%TYPE
                    ,io_job_id           IN OUT employees.job_id%TYPE
                    ,io_salary           IN OUT employees.salary%TYPE
                    ,io_commission_pct   IN OUT employees.commission_pct%TYPE
                    ,io_manager_id       IN OUT employees.manager_id%TYPE
                    ,io_department_id    IN OUT employees.department_id%TYPE
                    ,in_wait             INTEGER) IS
    l_status PLS_INTEGER;
    co_dflt_pipe_name CONSTANT STRING(30 CHAR) := 'MyPipe';
    co_ok CONSTANT PLS_INTEGER := 1;
  BEGIN
    -- Receive next message and unpack for each column.
    l_status := SYS.dbms_pipe.receive_message(pipename => co_dflt_pipe_name
                                             ,timeout => in_wait);

    IF l_status = co_ok THEN
      SYS.dbms_pipe.unpack_message (io_first_name);
      SYS.dbms_pipe.unpack_message (io_last_name);
      SYS.dbms_pipe.unpack_message (io_email);
      SYS.dbms_pipe.unpack_message (io_phone_number);
      SYS.dbms_pipe.unpack_message (io_hire_date);
      SYS.dbms_pipe.unpack_message (io_job_id);
      SYS.dbms_pipe.unpack_message (io_salary);
      SYS.dbms_pipe.unpack_message (io_commission_pct);
      SYS.dbms_pipe.unpack_message (io_manager_id);
      SYS.dbms_pipe.unpack_message (io_department_id);
    END IF;
  END rcv_emp;
END employee_up;
/
```

#### EXAMPLE (GOOD)

```

CREATE OR REPLACE PACKAGE BODY employee_up IS
  PROCEDURE rcv_emp (OUT_first_name      OUT employees.first_name%TYPE
                    ,OUT_last_name      OUT employees.last_name%TYPE
                    ,OUT_email          OUT employees.email%TYPE
                    ,OUT_phone_number    OUT employees.phone_number%TYPE
                    ,OUT_hire_date       OUT employees.hire_date%TYPE
                    ,OUT_job_id          OUT employees.job_id%TYPE
                    ,OUT_salary          OUT employees.salary%TYPE
                    ,OUT_commission_pct  OUT employees.commission_pct%TYPE
                    ,OUT_manager_id      OUT employees.manager_id%TYPE
                    ,OUT_department_id   OUT employees.department_id%TYPE
                    ,in_wait            IN      INTEGER) IS
    l_status PLS_INTEGER;
    co_dflt_pipe_name CONSTANT STRING(30 CHAR) := 'MyPipe';
    co_ok CONSTANT PLS_INTEGER := 1;
  BEGIN
    -- Receive next message and unpack for each column.
    l_status := SYS.dbms_pipe.receive_message(pipename => co_dflt_pipe_name
                                             ,timeout => in_wait);

    IF l_status = co_ok THEN
      SYS.dbms_pipe.unpack_message (out_first_name);
      SYS.dbms_pipe.unpack_message (out_last_name);
      SYS.dbms_pipe.unpack_message (out_email);
      SYS.dbms_pipe.unpack_message (out_phone_number);
      SYS.dbms_pipe.unpack_message (out_hire_date);
      SYS.dbms_pipe.unpack_message (out_job_id);
      SYS.dbms_pipe.unpack_message (out_salary);
      SYS.dbms_pipe.unpack_message (out_commission_pct);
      SYS.dbms_pipe.unpack_message (out_manager_id);
      SYS.dbms_pipe.unpack_message (out_department_id);
    END IF;
  END rcv_emp;
END employee_up;
/

```

## Procedures

### G-7310: Avoid standalone procedures – put your procedures in packages.



Minor

Maintainability

#### REASON

Use packages to structure your code, combine procedures and functions which belong together.

Package bodies may be changed and compiled without invalidating other packages. This is major advantage compared to standalone procedures and functions.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PROCEDURE my_procedure IS
BEGIN
    NULL;
END my_procedure;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE my_package IS
    PROCEDURE my_procedure;
END my_package;
/

CREATE OR REPLACE PACKAGE BODY my_package IS
    PROCEDURE my_procedure IS
    BEGIN
        NULL;
    END my_procedure;
END my_package;
/
```

## G-7320: Avoid using RETURN statements in a PROCEDURE.

### Major

Maintainability, Testability

#### REASON

Use of the `RETURN` statement is legal within a procedure in PL/SQL, but it is very similar to a `GOTO`, which means you end up with poorly structured code that is hard to debug and maintain.

A good general rule to follow as you write your PL/SQL programs is "one way in and one way out". In other words, there should be just one way to enter or call a program, and there should be one way out, one exit path from a program (or loop) on successful termination. By following this rule, you end up with code that is much easier to trace, debug, and maintain.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_procedure IS
    l_idx SIMPLE_INTEGER := 1;
    co_modulo CONSTANT SIMPLE_INTEGER := 7;
  BEGIN
    <<mod7_loop>>
    LOOP
      IF MOD(l_idx,co_modulo) = 0 THEN
        RETURN;
      END IF;

      l_idx := l_idx + 1;
    END LOOP mod7_loop;
  END my_procedure;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  PROCEDURE my_procedure IS
    l_idx SIMPLE_INTEGER := 1;
    co_modulo CONSTANT SIMPLE_INTEGER := 7;
  BEGIN
    <<mod7_loop>>
    LOOP
      EXIT mod7_loop WHEN MOD(l_idx,co_modulo) = 0;

      l_idx := l_idx + 1;
    END LOOP mod7_loop;
  END my_procedure;
END my_package;
/
```

## Functions

### G-7410: Avoid standalone functions – put your functions in packages.



Minor

Maintainability

#### REASON

Use packages to structure your code, combine procedures and functions which belong together.

Package bodies may be changed and compiled without invalidating other packages. This is major advantage compared to standalone procedures and functions.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE FUNCTION my_function RETURN VARCHAR2 IS
BEGIN
    RETURN NULL;
END my_function;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
    FUNCTION my_function RETURN VARCHAR2 IS
    BEGIN
        RETURN NULL;
    END my_function;
END my_package;
/
```



## G-7420: Always make the RETURN statement the last statement of your function.

### Major

Maintainability

#### REASON

The reader expects the RETURN statement to be the last statement of a function.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function (in_from IN PLS_INTEGER
                        , in_to   IN PLS_INTEGER) RETURN PLS_INTEGER IS
    l_ret PLS_INTEGER;
  BEGIN
    l_ret := in_from;
    <<for_loop>>
    FOR i IN in_from .. in_to
    LOOP
      l_ret := l_ret + i;
      IF i = in_to THEN
        RETURN l_ret;
      END IF;
    END LOOP for_loop;
  END my_function;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function (in_from IN PLS_INTEGER
                        , in_to   IN PLS_INTEGER) RETURN PLS_INTEGER IS
    l_ret PLS_INTEGER;
  BEGIN
    l_ret := in_from;
    <<for_loop>>
    FOR i IN in_from .. in_to
    LOOP
      l_ret := l_ret + i;
    END LOOP for_loop;
    RETURN l_ret;
  END my_function;
END my_package;
/
```

**G-7430: Try to use no more than one RETURN statement within a function.**

**⚠ Major**

Will have a medium/potential impact on the maintenance cost. Maintainability, Testability

**REASON**

A function should have a single point of entry as well as a single exit-point.

**EXAMPLE (BAD)**

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function (in_value IN PLS_INTEGER) RETURN BOOLEAN IS
    co_yes CONSTANT PLS_INTEGER := 1;
  BEGIN
    IF in_value = co_yes THEN
      RETURN TRUE;
    ELSE
      RETURN FALSE;
    END IF;
  END my_function;
END my_package;
/
```

**EXAMPLE (BETTER)**

```
CREATE OR REPLACE PACKAGE BODY my_package IS FUNCTION my_function (in_value IN PLS_INTEGER) RETURN
BOOLEAN IS co_yes CONSTANT PLS_INTEGER := 1; l_ret BOOLEAN; BEGIN IF in_value = co_yes THEN l_ret := TRUE; ELSE
l_ret := FALSE; END IF;
```

```
RETURN l_ret;
```

```
END my_function; END my_package; /
```

**EXAMPLE (GOOD)**

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function (in_value IN PLS_INTEGER) RETURN BOOLEAN IS
    co_yes CONSTANT PLS_INTEGER := 1;
  BEGIN
    RETURN in_value = co_yes;
  END my_function;
END my_package;
/
```

## G-7440: Never use OUT parameters to return values from a function.

### Major

Reusability

#### REASON

A function should return all its data through the RETURN clause. Having an OUT parameter prohibits usage of a function within SQL statements.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function (out_date OUT DATE) RETURN BOOLEAN IS
  BEGIN
    out_date := SYSDATE;
    RETURN TRUE;
  END my_function;
END my_package;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function RETURN DATE IS
  BEGIN
    RETURN SYSDATE;
  END my_function;
END my_package;
/
```

**G-7450: Never return a NULL value from a BOOLEAN function.**

 **Major**

Reliability, Testability

**REASON**

If a boolean function returns null, the caller has to deal with it. This makes the usage cumbersome and more error-prone.

**EXAMPLE (BAD)**

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function RETURN BOOLEAN IS
  BEGIN
    RETURN NULL;
  END my_function;
END my_package;
/
```

**EXAMPLE (GOOD)**

```
CREATE OR REPLACE PACKAGE BODY my_package IS
  FUNCTION my_function RETURN BOOLEAN IS
  BEGIN
    RETURN TRUE;
  END my_function;
END my_package;
/
```

**G-7460: Try to define your packaged/standalone function deterministic if appropriate.**

**⚠ Major**

Efficiency

**REASON**

A deterministic function (always return same result for identical parameters) which is defined to be deterministic will be executed once per different parameter within a SQL statement whereas if the function is not defined to be deterministic it is executed once per result row.

**EXAMPLE (BAD)**

```
CREATE OR REPLACE PACKAGE department_api IS
  FUNCTION name_by_id (in_department_id IN departments.department_id%TYPE)
    RETURN departments.department_name%TYPE;
END department_api;
/
```

**EXAMPLE (GOOD)**

```
CREATE OR REPLACE PACKAGE department_api IS
  FUNCTION name_by_id (in_department_id IN departments.department_id%TYPE)
    RETURN departments.department_name%TYPE DETERMINISTIC;
END department_api;
/
```

## Oracle Supplied Packages

**G-7510: Always prefix ORACLE supplied packages with owner schema name.**

### Major

Security

#### REASON

The signature of oracle-supplied packages is well known and therefore it is quite easy to provide packages with the same name as those from oracle doing something completely different without you noticing it.

#### EXAMPLE (BAD)

```
DECLARE
    co_hello_world CONSTANT STRING(30 CHAR) := 'Hello World';
BEGIN
    dbms_output.put_line(co_hello_world);
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
    co_hello_world CONSTANT STRING(30 CHAR) := 'Hello World';
BEGIN
    sys.dbms_output.put_line(co_hello_world);
END;
/
```

## Object Types

There are no object type-specific recommendations to be defined at the time of writing.

## Triggers

### G-7710: Avoid cascading triggers.

#### Major

Maintainability, Testability

#### REASON

Having triggers that act on other tables in a way that causes triggers on that table to fire lead to obscure behavior.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE TRIGGER dept_br_u
BEFORE UPDATE ON departments FOR EACH ROW
BEGIN
    INSERT INTO departments_hist (department_id
                                , department_name
                                , manager_id
                                , location_id
                                , modification_date)
        VALUES (:OLD.department_id
                , :OLD.department_name
                , :OLD.manager_id
                , :OLD.location_id
                , SYSDATE);
END;
/
CREATE OR REPLACE TRIGGER dept_hist_br_i
BEFORE INSERT ON departments_hist FOR EACH ROW
BEGIN
    INSERT INTO departments_log (department_id
                                , department_name
                                , modification_date)
        VALUES (:NEW.department_id
                , :NEW.department_name
                , SYSDATE);
END;
/
```

#### EXAMPLE (GOOD)



```
CREATE OR REPLACE TRIGGER dept_br_u
BEFORE UPDATE ON departments FOR EACH ROW
BEGIN
    INSERT INTO departments_hist (department_id
                                ,department_name
                                ,manager_id
                                ,location_id
                                ,modification_date)
        VALUES (:OLD.department_id
                ,:OLD.department_name
                ,:OLD.manager_id
                ,:OLD.location_id
                ,SYSDATE);

    INSERT INTO departments_log (department_id
                                ,department_name
                                ,modification_date)
        VALUES (:OLD.department_id
                ,:OLD.department_name
                ,SYSDATE);

END;
/
```

## Sequences

**G-7810: Never use SQL inside PL/SQL to read sequence numbers (or SYSDATE).**

### Major

Efficiency, Maintainability

#### REASON

Since ORACLE 11g it is no longer needed to use a SELECT statement to read a sequence (which would imply a context switch).

#### EXAMPLE (BAD)

```
DECLARE
  l_sequence_number employees.employee_id%type;
BEGIN
  SELECT employees_seq.NEXTVAL
     INTO l_sequence_number
    FROM DUAL;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
  l_sequence_number employees.employee_id%type;
BEGIN
  l_sequence_number := employees_seq.NEXTVAL;
END;
/
```

# Patterns

## Checking the Number of Rows

**G-8110: Never use SELECT COUNT(\*) if you are only interested in the existence of a row.**

### Major

Efficiency

#### REASON

If you do a `SELECT count()` all rows will be read according to the `WHERE` clause, even if only the availability of data is of interest. For this we have a big performance overhead. If we do a `SELECT count() ... WHERE ROWNUM = 1` there is also a overhead as there will be two communications between the PL/SQL and the SQL engine. See the following example for a better solution.

#### EXAMPLE (BAD)

```
DECLARE
  l_count PLS_INTEGER;
  co_zero  CONSTANT SIMPLE_INTEGER := 0;
  co_salary CONSTANT employees.salary%TYPE := 5000;
BEGIN
  SELECT count(*)
    INTO l_count
  FROM employees
 WHERE salary < co_salary;
  IF l_count > co_zero THEN
    <<emp_loop>>
    FOR r_emp IN (SELECT employee_id
                  FROM employees)
      LOOP
        IF r_emp.salary < co_salary THEN
          my_package.my_proc(in_employee_id => r_emp.employee_id);
        END IF;
      END LOOP emp_loop;
  END IF;
END;
/
```

#### EXAMPLE (GOOD)

```
DECLARE
  co_salary CONSTANT employees.salary%TYPE := 5000;
BEGIN
  <<emp_loop>>
  FOR r_emp IN (SELECT e1.employee_id
                FROM employees e1
                WHERE EXISTS(SELECT e2.salary
                             FROM employees e2
                             WHERE e2.salary < co_salary))
    LOOP
      my_package.my_proc(in_employee_id => r_emp.employee_id);
    END LOOP emp_loop;
END;
/
```

## G-8120: Never check existence of a row to decide whether to create it or not.

### Major

Efficiency, Reliability

#### REASON

The result of an existence check is a snapshot of the current situation. You never know whether in the time between the check and the (insert) action someone else has decided to create a row with the values you checked. Therefore, you should only rely on constraints when it comes to prevention of duplicate records.

#### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  PROCEDURE ins (in_r_department IN departments%ROWTYPE) IS
    l_count PLS_INTEGER;
  BEGIN
    SELECT count(*)
      INTO l_count
    FROM departments
    WHERE department_id = in_r_department.department_id;

    IF l_count = 0 THEN
      INSERT INTO departments
        VALUES in_r_department;
    END IF;
  END ins;
END department_api;
/
```

#### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  PROCEDURE ins (in_r_department IN departments%ROWTYPE) IS
  BEGIN
    INSERT INTO departments
      VALUES in_r_department;
  EXCEPTION
    WHEN dup_val_on_index THEN NULL; -- handle exception
  END ins;
END department_api;
/
```

## Access objects of foreign application schemas

**G-8210: Always use synonyms when accessing objects of another application schema.**

### Major

Changeability, Maintainability

#### REASON

If a connection is needed to a table that is placed in a foreign schema, using synonyms is a good choice. If there are structural changes to that table (e.g. the table name changes or the table changes into another schema) only the synonym has to be changed no changes to the package are needed (single point of change). If you only have read access for a table inside another schema, or there is another reason that does not allow you to change data in this table, you can switch the synonym to a table in your own schema. This is also good practice for testers working on test systems.

#### EXAMPLE (BAD)

```
DECLARE
  l_product_name oe.products.product_name%TYPE;
  co_price CONSTANT oe.products@list_price%TYPE := 1000;
BEGIN
  SELECT p.product_name
    INTO l_product_name
    FROM oe.products p
   WHERE list_price > co_price;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    NULL; -- handle_no_data_found;
  WHEN TOO_MANY_ROWS THEN
    NULL; -- handle_too_many_rows;
END;
/
```

#### EXAMPLE (GOOD)

```
CREATE SYNONYM oe_products FOR oe.products;

DECLARE
  l_product_name oe_products.product_name%TYPE;
  co_price CONSTANT oe_products.list_price%TYPE := 1000;
BEGIN
  SELECT p.product_name
    INTO l_product_name
    FROM oe_products p
   WHERE list_price > co_price;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    NULL; -- handle_no_data_found;
  WHEN TOO_MANY_ROWS THEN
    NULL; -- handle_too_many_rows;
END;
/
```

## Validating input parameter size

**G-8310: Always validate input parameter size by assigning the parameter to a size limited variable in the declaration section of program unit.**



Minor

Maintainability, Reliability, Reusability, Testability

### REASON

This technique raises an error (value\_error) which may not be handled in the called program unit. This is the right way to do it, as the error is not within this unit but when calling it, so the caller should handle the error.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  FUNCTION dept_by_name (in_dept_name IN departments.department_name%TYPE)
    RETURN departments%ROWTYPE IS
    l_return departments%rowtype;
  BEGIN
    IF in_dept_name IS NULL
      OR LENGTH(in_dept_name) > 20
    THEN
      RAISE err.e_param_to_large;
    END IF;
    -- get the department by name
    SELECT *
      FROM departments
     WHERE department_name = in_dept_name;

    RETURN l_return;
  END dept_by_name;
END department_api;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY department_api IS
  FUNCTION dept_by_name (in_dept_name IN departments.department_name%TYPE)
    RETURN departments%ROWTYPE IS
    l_dept_name departments.department_name%TYPE NOT NULL := in_dept_name;
    l_return departments%rowtype;
  BEGIN
    -- get the department by name
    SELECT *
      FROM departments
     WHERE department_name = l_dept_name;

    RETURN l_return;
  END dept_by_name;
END department_api;
/
```

### FUNCTION CALL

```
...  
    r_deparment := department_api.dept_by_name('Far to long name of a department');  
...  
EXCEPTION  
    WHEN VALUE_ERROR THEN ...
```

## Ensure single execution at a time of a program unit

**G-8410: Always use application locks to ensure a program unit is only running once at a given time.**



Minor

Efficiency, Reliability

### REASON

This technique allows us to have locks across transactions as well as a proven way to clean up at the end of the session.

The alternative using a table where a “Lock-Row” is stored has the disadvantage that in case of an error a proper cleanup has to be done to “unlock” the program unit.

### EXAMPLE (BAD)

```
-- Bad
/* Example */
CREATE OR REPLACE PACKAGE BODY lock_up IS
    -- manage locks in a dedicated table created as follows:
    -- CREATE TABLE app_locks (
    --     lock_name VARCHAR2(128 CHAR) NOT NULL primary key
    -- );

    PROCEDURE request_lock (in_lock_name IN VARCHAR2) IS
    BEGIN
        -- raises dup_val_on_index
        INSERT INTO app_locks (lock_name) VALUES (in_lock_name);
    END request_lock;

    PROCEDURE release_lock(in_lock_name IN VARCHAR2) IS
    BEGIN
        DELETE FROM app_locks WHERE lock_name = in_lock_name;
    END release_lock;
END lock_up;
/

/* Call bad example */
DECLARE
    co_lock_name CONSTANT VARCHAR2(30 CHAR) := 'APPLICATION_LOCK';
BEGIN
    lock_up.request_lock(in_lock_name => co_lock_name);
    -- processing
    lock_up.release_lock(in_lock_name => co_lock_name);
EXCEPTION
    WHEN OTHERS THEN
        -- log error
        lock_up.release_lock(in_lock_name => co_lock_name);
        RAISE;
END;
/
```

### EXAMPLE (GOOD)



```

/* Example */
CREATE OR REPLACE PACKAGE BODY lock_up IS
  FUNCTION request_lock(
    in_lock_name          IN VARCHAR2,
    in_release_on_commit IN BOOLEAN := FALSE)
  RETURN VARCHAR2 IS
    l_lock_handle VARCHAR2(128 CHAR);
  BEGIN
    sys.dbms_lock.allocate_unique(
      lockname      => in_lock_name,
      lockhandle    => l_lock_handle,
      expiration_secs => constants_up.co_one_week
    );
    IF sys.dbms_lock.request(
      lockhandle    => l_lock_handle,
      lockmode      => sys.dbms_lock.x_mode,
      timeout       => sys.dbms_lock.maxwait,
      release_on_commit => COALESCE(in_release_on_commit, FALSE)
    ) > 0
    THEN
      RAISE err.e_lock_request_failed;
    END IF;
    RETURN l_lock_handle;
  END request_lock;

  PROCEDURE release_lock(in_lock_handle IN VARCHAR2) IS
  BEGIN
    IF sys.dbms_lock.release(lockhandle => in_lock_handle) > 0 THEN
      RAISE err.e_lock_request_failed;
    END IF;
  END release_lock;
END lock_up;
/

/* Call good example */
DECLARE
  l_handle VARCHAR2(128 CHAR);
  co_lock_name CONSTANT VARCHAR2(30 CHAR) := 'APPLICATION_LOCK';
BEGIN
  l_handle := lock_up.request_lock(in_lock_name => co_lock_name);
  -- processing
  lock_up.release_lock(in_lock_handle => l_handle);
EXCEPTION
  WHEN OTHERS THEN
    -- log error
    lock_up.release_lock(in_lock_handle => l_handle);
    RAISE;
END;
/

```

## Use dbms\_application\_info package to follow progress of a process

**G-8510: Always use dbms\_application\_info to track program process transiently.**



Minor

Efficiency, Reliability

### REASON

This technique allows us to view progress of a process without having to persistently write log data in either a table or a file. The information is accessible through the `V$SESSION` view.

### EXAMPLE (BAD)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
  PROCEDURE process_emps IS
  BEGIN
    <<employees>>
    FOR emp_rec IN (SELECT employee_id
                     FROM employees
                     ORDER BY employee_id)

    LOOP
      NULL; -- some processing
    END LOOP employees;
  END process_emps;
END employee_api;
/
```

### EXAMPLE (GOOD)

```
CREATE OR REPLACE PACKAGE BODY employee_api IS
  PROCEDURE process_emps IS
  BEGIN
    SYS.DBMS_APPLICATION_INFO.SET_MODULE(module_name => $$PLSQL_UNIT
                                          ,action_name => 'Init');

    <<employees>>
    FOR emp_rec IN (SELECT employee_id
                     FROM employees
                     ORDER BY employee_id)

    LOOP
      SYS.DBMS_APPLICATION_INFO.SET_ACTION('Processing ' || emp_rec.employee_id);
    END LOOP employees;
  end process_emps;
END employee_api;
/
```

# Complexity Analysis

Using software metrics like complexity analysis will guide you towards maintainable and testable pieces of code by reducing the complexity and splitting the code into smaller chunks.

## Halstead Metrics

### Calculation

First, we need to compute the following numbers, given the program:

- $n_1$  = the number of distinct operators
- $n_2$  = the number of distinct operands
- $N_1$  = the total number of operators
- $N_2$  = the total number of operands

From these numbers, five measures can be calculated:

- Program length:  
$$N = N_1 + N_2$$
- Program vocabulary:  
$$n = n_1 + n_2$$
- Volume:  
$$V = N \cdot \log_2 n$$
- Difficulty:  
$$D = \frac{n_1}{2} \cdot \frac{N_2}{n_2}$$
- Effort:  
$$E = D \cdot V$$

The difficulty measure

$D$  is related to the difficulty of the program to write or understand, e.g. when doing code review.

The volume measure

$V$  describes the size of the implementation of an algorithm.

## McCabe's Cyclomatic Complexity

### Description

Cyclomatic complexity (or conditional complexity) is a software metric used to measure the complexity of a program. It directly measures the number of linearly independent paths through a program's source code.

Cyclomatic complexity is computed using the control flow graph of the program: the nodes of the graph correspond to indivisible groups of commands of a program, and a directed edge connects two nodes if the second command might be executed immediately after the first command. Cyclomatic complexity may also be applied to individual functions, modules, methods or classes within a program.

The cyclomatic complexity of a section of source code is the count of the number of linearly independent paths through

the source code. For instance, if the source code contains no decision points, such as IF statements or FOR loops, the complexity would be 1, since there is only a single path through the code. If the code has a single IF statement containing a single condition there would be two paths through the code, one path where the IF statement is evaluated as TRUE and one path where the IF statement is evaluated as FALSE.

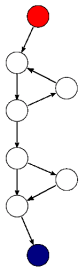
## Calculation

Mathematically, the cyclomatic complexity of a structured program is defined with reference to a directed graph containing the basic blocks of the program, with an edge between two basic blocks if control may pass from the first to the second (the control flow graph of the program). The complexity is then defined as:

$$M = E - N + 2P$$

where

- $M$  = cyclomatic complexity
- $E$  = the number of edges of the graph
- $N$  = the number of nodes of the graph
- $P$  = the number of connected components.



Take, for example, a control flow graph of a simple program. The program begins executing at the red node, then enters a loop (group of three nodes immediately below the red node). On exiting the loop, there is a conditional statement (group below the loop), and finally the program exits at the blue node.

For this graph,

$E = 9$ ,

$N = 8$  and

$P = 1$ , so the cyclomatic complexity of the program is

3.

```
BEGIN
  FOR i IN 1..3
    LOOP
      dbms_output.put_line('in loop');
    END LOOP;
    --
    IF 1 = 1
      THEN
        dbms_output.put_line('yes');
      END IF;
    --
    dbms_output.put_line('end');
  END;
/
```

For a single program (or subroutine or method),  $P$  is always equal to 1. Cyclomatic complexity may, however, be applied to several such programs or subprograms at the same time (e.g., to all of the methods in a class), and in these cases  $P$  will be equal to the number of programs in question, as each subprogram will appear as a disconnected subset of the graph.

It can be shown that the cyclomatic complexity of any structured program with only one entrance point and one exit point is equal to the number of decision points (i.e., 'if' statements or conditional loops) contained in that program plus one.

Cyclomatic complexity may be extended to a program with multiple exit points; in this case it is equal to:

$$\pi = s + 2$$

Where

- $\pi$  is the number of decision points in the program, and
- $s$  is the number of exit points.

# Code Reviews

Code reviews check the results of software engineering. According to IEEE-Norm 729, a review is a more or less planned and structured analysis and evaluation process. Here we distinguish between code review and architect review.

To perform a code review means that after or during the development one or more reviewer proof-reads the code to find potential errors, potential areas for simplification, or test cases. A code review is a very good opportunity to save costs by fixing issues before the testing phase.

What can a code-review be good for?

- Code quality
- Code clarity and maintainability
- Quality of the overall architecture
- Quality of the documentation
- Quality of the interface specification

For an effective review, the following factors must be considered:

- Definition of clear goals.
- Choice of a suitable person with constructive critical faculties.
- Psychological aspects.
- Selection of the right review techniques.
- Support of the review process from the management.
- Existence of a culture of learning and process optimization.

Requirements for the reviewer:

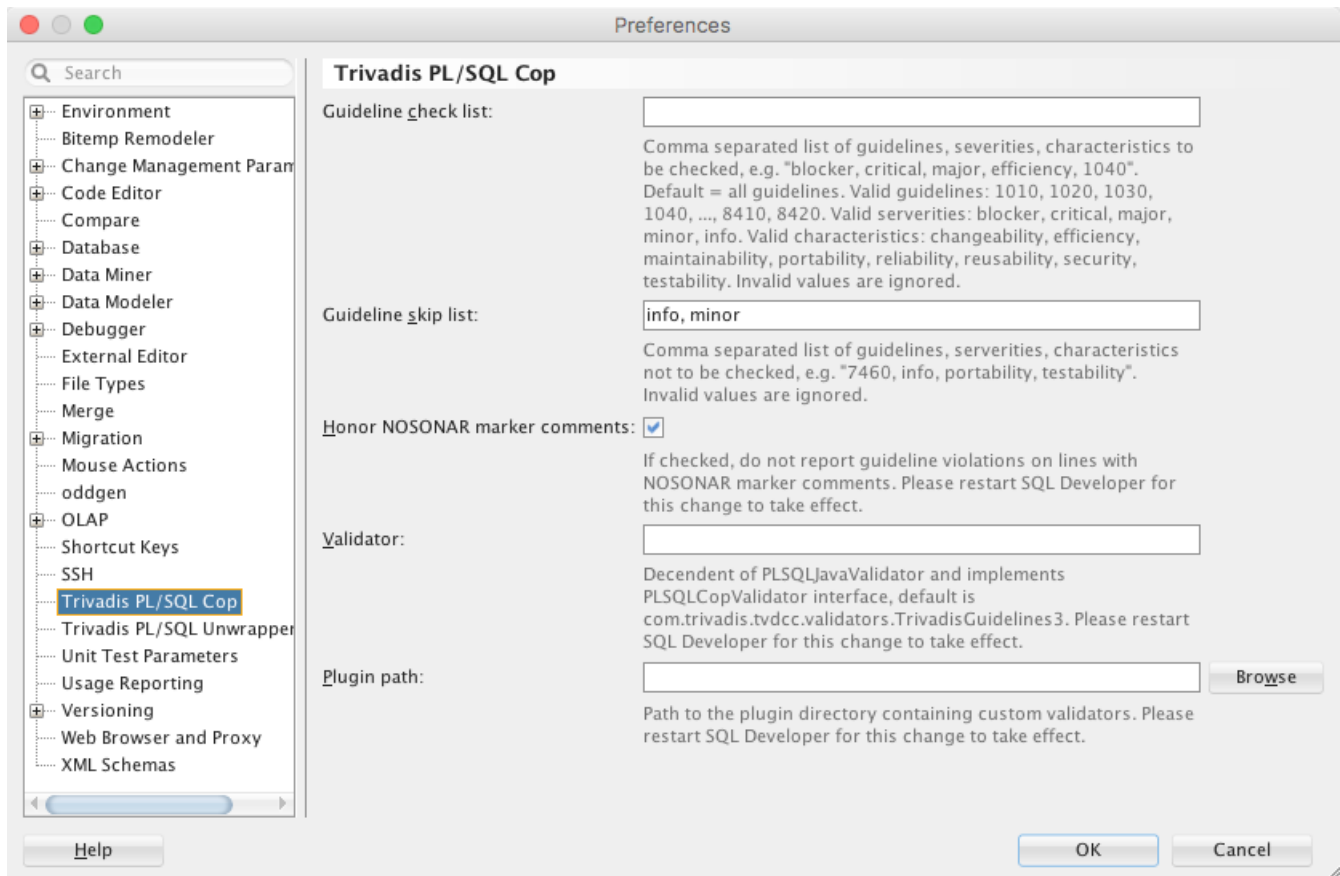
- He must not be the owner of the code.
- Code reviews may be unpleasant for the developer, as he could fear that his code will be criticized. If the critic is not considerate, the code writer will build up rejection and resistance against code reviews.

# Tool Support

## Development

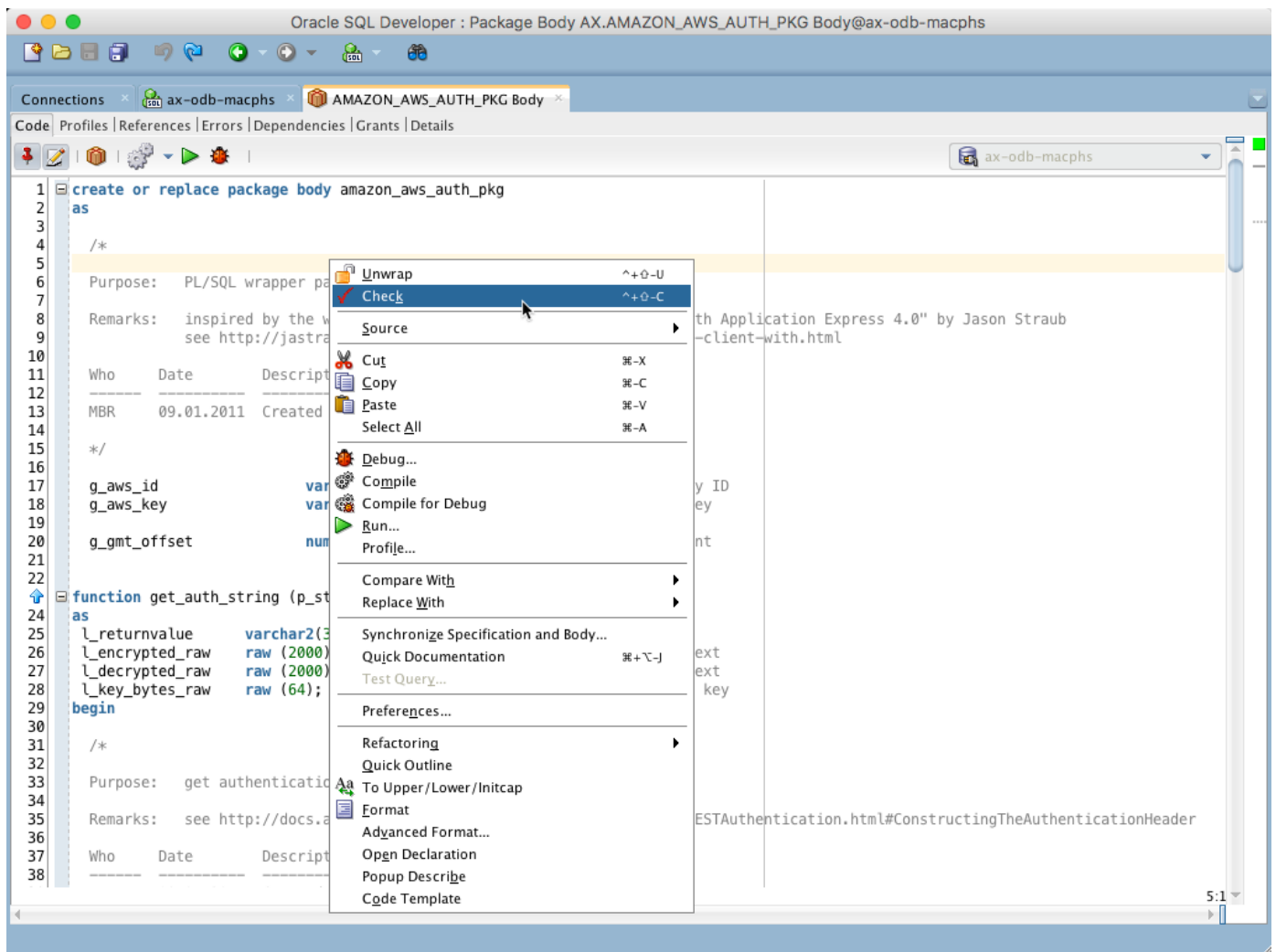
Trivadis offers a cost-free extension to ORACLE SQL Developer to test compliance with this coding guideline. The extension may be parameterized to your preferred set of rules and allows checking this set against a program unit.

### Setting the preferences



There is an include list as well as an exclude list to define which rules to be checked or ignored.

### Activate PLSQL Cop using context menu



The result of the checking process is a list of violations with direct links to the place in the code as well as software metrics like:

- Cyclomatic complexity
- Halstead volume
- Maintainability Index
- Number of lines of code
- Number of comment lines
- Issue Overview

This statistics are gathered for each program unit in the reviewed code.

## Software metrics



Oracle SQL Developer

Trivadis PL/SQL Cop

IssuesReport

Trivadis PL/SQL Cop Version 2.0.0.2244 for Trivadis PL/SQL & SQL Coding Guidelines Version 3.2

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CH-8152 Glattbrugg (Zurich)  
[www.trivadis.com](http://www.trivadis.com)

Parameters

check (all guidelines)  
skip info, minor  
nosonar yes – honor NOSONAR marker comments (do not report guideline violations on lines with NOSONAR marker comments)  
validator (default validator – com.trivadis.tvdcc.validators.TrivadisGuidelines3)  
plugin-path (no directory configured)

SQL Developer Editor

Title Package Body AX.AMAZON\_AWS\_AUTH\_PKG@ax-odb-macphs  
Date/time 2017-01-29 18:21:16

Metrics

Number of bytes 4,962  
Number of lines (LOC) 246  
Number of comment lines 111  
Number of blank lines 58  
Number of net lines 77  
Number of commands 1  
Number of statements (PL/SQL) 22  
Max. cyclomatic complexity 2 ( < 11 11.50 > 50 )  
Max. Halstead volume 333 ( < 1001 1001..3000 > 3000 )  
Min. maintainability index (MI) 141 ( > 84 64.84 < 64 )  
Avg cyclomatic complexity 1 ( < 11 11.50 > 50 )  
Avg Halstead volume 77 ( < 1001 1001..3000 > 3000 )  
Avg maintainability index (MI) 161 ( > 84 64.84 < 64 )  
Number of issues 11  
Number of warnings 11  
Number of errors 0

PL/SQL Units

| PL/SQL Unit                         | Line | # Lines | # Comment lines | # Blank lines | # Net lines | # Stmts | Cyclomatic complexity | Halstead volume | Maintainability index |
|-------------------------------------|------|---------|-----------------|---------------|-------------|---------|-----------------------|-----------------|-----------------------|
| amazon_aws_auth_pkg.get_date_string | 105  | 25      | 11              | 4             | 10          | 6       | 2                     | 261             | 142                   |
| amazon_aws_auth_pkg.get_auth_string | 29   | 26      | 12              | 7             | 8           | 6       | 1                     | 333             | 141                   |
| amazon_aws_auth_pkg.init            | 224  | 19      | 11              | 3             | 5           | 3       | 1                     | 39              | 164                   |
| amazon_aws_auth_pkg.get_epoch       | 135  | 19      | 11              | 4             | 4           | 2       | 1                     | 94              | 160                   |
| amazon_aws_auth_pkg.get_signature   | 60   | 17      | 11              | 3             | 3           | 1       | 1                     | 30              | 171                   |
| amazon_aws_auth_pkg.get_aws_id      | 81   | 17      | 11              | 3             | 3           | 1       | 1                     | 6               | 180                   |
| amazon_aws_auth_pkg.set_aws_id      | 158  | 18      | 11              | 4             | 3           | 1       | 1                     | 10              | 174                   |
| amazon_aws_auth_pkg.set_aws_key     | 180  | 17      | 11              | 3             | 3           | 1       | 1                     | 10              | 177                   |
| amazon_aws_auth_pkg.set_gmt_offset  | 201  | 17      | 11              | 3             | 3           | 1       | 1                     | 10              | 177                   |

Issue Overview

| # | %     | Severity | Characteristics | Message                                                                                     |
|---|-------|----------|-----------------|---------------------------------------------------------------------------------------------|
| 6 | 54.5% | Major    | Security        | G-7510: Always prefix ORACLE supplied packages with owner schema name.                      |
| 5 | 45.5% | Major    | Efficiency      | G-7460: Try to define your packaged/standalone function to be deterministic if appropriate. |

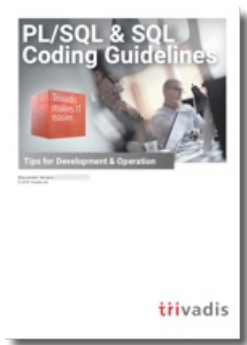
Open

# Appendix

## A - PL/SQL & SQL Coding Guidelines as PDF

These guidelines are primarily produced in [HTML](#) using [Material for MkDocs](#).

However, we provide these guidelines also as [PDF](#) produced by [wkhtmltopdf](#).



The formatting is not perfect, but it should be adequate for those who want to work with offline documents.

## B - Mapping new guidelines to prior versions

| Old Id | New Id | Text                                                                 | Severity | Change-ability | Efficiency | Maintain-ability | Portability |
|--------|--------|----------------------------------------------------------------------|----------|----------------|------------|------------------|-------------|
| 1      | 1010   | Try to label your sub blocks.                                        | Minor    |                |            | X                |             |
| 2      | 1020   | Always have a matching loop or block label.                          | Minor    |                |            | X                |             |
| 3      | 1030   | Avoid defining variables that are not used.                          | Minor    |                | X          | X                |             |
| 4      | 1040   | Avoid dead code.                                                     | Minor    |                |            | X                |             |
| 5      | 1050   | Avoid using literals in your code.                                   | Minor    | X              |            |                  |             |
| 6      | 1060   | Avoid storing ROWIDs or UROWIDs in database tables.                  | Major    |                |            |                  |             |
| 7      | 1070   | Avoid nesting comment blocks.                                        | Minor    |                |            | X                |             |
| 8      | 2110   | Try to use anchored declarations for variables, constants and types. | Major    |                |            | X                |             |
| 9      | 2120   | Try to have a single location to define your types.                  | Minor    | X              |            |                  |             |
| 10     | 2130   | Try to use subtypes for constructs used often in your                | Minor    | X              |            |                  |             |

|     |      |                                                                                         |          |  |   |   |   |
|-----|------|-----------------------------------------------------------------------------------------|----------|--|---|---|---|
|     |      | code.                                                                                   |          |  |   |   |   |
| 11  | 2140 | Never initialize variables with NULL.                                                   | Minor    |  |   | X |   |
| 12  | 2150 | Avoid comparisons with NULL value, consider using IS [NOT] NULL.                        | Blocker  |  |   |   | X |
| 13  | 2160 | Avoid initializing variables using functions in the declaration section.                | Critical |  |   |   |   |
| 14  | 2170 | Never overload variables.                                                               | Major    |  |   |   |   |
| 15  | 2180 | Never use quoted identifiers.                                                           | Major    |  |   | X |   |
| 16  | 2185 | Avoid using overly short names for explicitly or implicitly declared identifiers.       | Minor    |  |   | X |   |
| 17  | 2190 | Avoid the use of ROWID or UROWID.                                                       | Major    |  |   |   | X |
| 18  | 2210 | Avoid declaring NUMBER variables or subtypes with no precision.                         | Minor    |  | X |   |   |
| 19  | 2220 | Try to use PLS_INTEGER instead of NUMBER for arithmetic operations with integer values. | Minor    |  | X |   |   |
| n/a | 2230 | Try to use SIMPLE_INTEGER datatype when appropriate.                                    | Minor    |  | X |   |   |
| 20  | 2310 | Avoid using CHAR data type.                                                             | Major    |  |   |   |   |
| 21  | 2320 | Avoid using VARCHAR data type.                                                          | Major    |  |   |   | X |
| 22  | 2330 | Never use zero-length strings to substitute NULL.                                       | Major    |  |   |   | X |
| 23  | 2340 | Always define your VARCHAR2 variables using CHAR SEMANTIC (if not defined anchored).    | Minor    |  |   |   |   |
| 24  | 2410 | Try to use boolean data type for values with dual meaning.                              | Minor    |  |   | X |   |
| 25  | 2510 | Avoid using the LONG and LONG RAW data types.                                           | Major    |  |   |   | X |
| 26  | 3110 | Always specify the target columns when coding an insert statement.                      | Major    |  |   | X |   |

|     |      |                                                                                                                       |          |   |   |   |   |
|-----|------|-----------------------------------------------------------------------------------------------------------------------|----------|---|---|---|---|
| 27  | 3120 | Always use table aliases when your SQL statement involves more than one source.                                       | Major    |   |   | X |   |
| 28  | 3130 | Try to use ANSI SQL-92 join syntax.                                                                                   | Minor    |   |   | X | X |
| 29  | 3140 | Try to use anchored records as targets for your cursors.                                                              | Major    |   |   | X |   |
| n/a | 3150 | Try to use identity columns for surrogate keys.                                                                       | Minor    |   |   | X |   |
| n/a | 3160 | Avoid virtual columns to be visible.                                                                                  | Major    |   |   | X |   |
| n/a | 3170 | Always use DEFAULT ON NULL declarations to assign default values to table columns if you refuse to store NULL values. | Major    |   |   |   |   |
| n/a | 3180 | Always specify column names instead of positional references in ORDER BY clauses.                                     | Major    | X |   |   |   |
| n/a | 3190 | Avoid using NATURAL JOIN.                                                                                             | Major    | X |   |   |   |
| 30  | 3210 | Always use BULK OPERATIONS (BULK COLLECT, FORALL) whenever you have to execute a DML statement more than 4 times.     | Major    |   | X |   |   |
| 31  | 4110 | Always use %NOTFOUND instead of NOT %FOUND to check whether a cursor returned data.                                   | Minor    |   |   | X |   |
| 32  | 4120 | Avoid using %NOTFOUND directly after the FETCH when working with BULK OPERATIONS and LIMIT clause.                    | Critical |   |   |   |   |
| 33  | 4130 | Always close locally opened cursors.                                                                                  | Major    |   | X |   |   |
| 34  | 4140 | Avoid executing any statements between a SQL operation and the usage of an implicit cursor attribute.                 | Major    |   |   |   |   |
| 35  | 4210 | Try to use CASE rather than an IF statement with multiple ELSIF paths.                                                | Major    |   |   | X |   |

|     |      |                                                                                                                  |          |   |   |   |   |
|-----|------|------------------------------------------------------------------------------------------------------------------|----------|---|---|---|---|
| 36  | 4220 | Try to use CASE rather than DECODE.                                                                              | Minor    |   |   | X | X |
| 37  | 4230 | Always use COALESCE instead of NVL, if parameter 2 of the NVL function is a function call or a SELECT statement. | Critical |   | X |   |   |
| 38  | 4240 | Always use CASE instead of NVL2 if parameter 2 or 3 of NVL2 is either a function call or a SELECT statement.     | Critical |   | X |   |   |
| 39  | 4310 | Never use GOTO statements in your code.                                                                          | Major    |   |   | X |   |
| 40  | 4320 | Always label your loops.                                                                                         | Minor    |   |   | X |   |
| 41  | 4330 | Always use a CURSOR FOR loop to process the complete cursor results unless you are using bulk operations.        | Minor    |   |   | X |   |
| 42  | 4340 | Always use a NUMERIC FOR loop to process a dense array.                                                          | Minor    |   |   | X |   |
| 43  | 4350 | Always use 1 as lower and COUNT() as upper bound when looping through a dense array.                             | Major    |   |   |   |   |
| 44  | 4360 | Always use a WHILE loop to process a loose array.                                                                | Minor    |   | X |   |   |
| 45  | 4370 | Avoid using EXIT to stop loop processing unless you are in a basic loop.                                         | Major    |   |   | X |   |
| 46  | 4375 | Always use EXIT WHEN instead of an IF statement to exit from a loop.                                             | Minor    |   |   | X |   |
| 47  | 4380 | Try to label your EXIT WHEN statements.                                                                          | Minor    |   |   | X |   |
| 48  | 4385 | Never use a cursor for loop to check whether a cursor returns data.                                              | Major    |   | X |   |   |
| 49  | 4390 | Avoid use of unreferenced FOR loop indexes.                                                                      | Major    |   | X |   |   |
| 50  | 4395 | Avoid hard-coded upper or lower bound values with FOR loops.                                                     | Minor    | X |   | X |   |
| n/a | 5010 | Try to use a error/logging                                                                                       | Critical |   |   |   |   |

|     |      |                                                                                                                         |          |   |   |   |  |
|-----|------|-------------------------------------------------------------------------------------------------------------------------|----------|---|---|---|--|
| n/a | 5010 | Try to use a error/logging framework for your application.                                                              | Critical |   |   |   |  |
| 51  | 5020 | Never handle unnamed exceptions using the error number.                                                                 | Critical |   |   | X |  |
| 52  | 5030 | Never assign predefined exception names to user defined exceptions.                                                     | Blocker  |   |   |   |  |
| 53  | 5040 | Avoid use of WHEN OTHERS clause in an exception section without any other specific handlers.                            | Major    |   |   |   |  |
| 54  | n/a  | Avoid use of EXCEPTION_INIT pragma for a 20nnn error.                                                                   | Major    |   |   |   |  |
| 55  | 5050 | Avoid use of the RAISE_APPLICATION_ERROR built-in procedure with a hard-coded 20nnn error number or hard-coded message. | Major    | X |   | X |  |
| 56  | 5060 | Avoid unhandled exceptions                                                                                              | Major    |   |   |   |  |
| 57  | 5070 | Avoid using Oracle predefined exceptions                                                                                | Critical |   |   |   |  |
| 58  | 6010 | Always use a character variable to execute dynamic SQL.                                                                 | Major    |   |   | X |  |
| 59  | 6020 | Try to use output bind arguments in the RETURNING INTO clause of dynamic DML statements rather than the USING clause.   | Minor    |   |   | X |  |
| 60  | 7110 | Try to use named notation when calling program units.                                                                   | Major    | X |   | X |  |
| 61  | 7120 | Always add the name of the program unit to its end keyword.                                                             | Minor    |   |   | X |  |
| 62  | 7130 | Always use parameters or pull in definitions rather than referencing external variables in a local program unit.        | Major    |   |   | X |  |
| 63  | 7140 | Always ensure that locally defined procedures or functions are referenced.                                              | Major    |   |   | X |  |
| 64  | 7150 | Try to remove unused                                                                                                    | Minor    |   | X | X |  |

|     |      |                                                                                                               |       |   |   |   |  |
|-----|------|---------------------------------------------------------------------------------------------------------------|-------|---|---|---|--|
|     |      | parameters.                                                                                                   |       |   |   |   |  |
| 65  | 7210 | Try to keep your packages small. Include only few procedures and functions that are used in the same context. | Minor |   | X | X |  |
| 66  | 7220 | Always use forward declaration for private functions and procedures.                                          | Minor | X |   |   |  |
| 67  | 7230 | Avoid declaring global variables public.                                                                      | Major |   |   |   |  |
| 68  | 7240 | Avoid using an IN OUT parameter as IN or OUT only.                                                            | Major |   | X | X |  |
| 69  | 7310 | Avoid standalone procedures – put your procedures in packages.                                                | Minor |   |   | X |  |
| 70  | 7320 | Avoid using RETURN statements in a PROCEDURE.                                                                 | Major |   |   | X |  |
| 71  | 7410 | Avoid standalone functions – put your functions in packages.                                                  | Minor |   |   | X |  |
| 73  | 7420 | Always make the RETURN statement the last statement of your function.                                         | Major |   |   | X |  |
| 72  | 7430 | Try to use no more than one RETURN statement within a function.                                               | Major |   |   | X |  |
| 74  | 7440 | Never use OUT parameters to return values from a function.                                                    | Major |   |   |   |  |
| 75  | 7450 | Never return a NULL value from a BOOLEAN function.                                                            | Major |   |   |   |  |
| n/a | 7460 | Try to define your packaged/standalone function to be deterministic if appropriate.                           | Major |   | X |   |  |
| 76  | 7510 | Always prefix ORACLE supplied packages with owner schema name.                                                | Major |   |   |   |  |
| 77  | 7710 | Avoid cascading triggers.                                                                                     | Major |   |   | X |  |
| n/a | 7810 | Do not use SQL inside PL/SQL to read sequence numbers (or SYSDATE)                                            | Major |   | X | X |  |
|     |      |                                                                                                               |       |   |   |   |  |

|     |      |                                                                                                                                        |       |   |   |   |  |
|-----|------|----------------------------------------------------------------------------------------------------------------------------------------|-------|---|---|---|--|
| 78  | 8110 | Never use SELECT COUNT(*) if you are only interested in the existence of a row.                                                        | Major |   | X |   |  |
| n/a | 8120 | Never check existence of a row to decide whether to create it or not.                                                                  | Major |   | X |   |  |
| 79  | 8210 | Always use synonyms when accessing objects of another application schema.                                                              | Major | X |   | X |  |
| n/a | 8310 | Always validate input parameter size by assigning the parameter to a size limited variable in the declaration section of program unit. | Minor |   |   | X |  |
| n/a | 8410 | Always use application locks to ensure a program unit only running once at a given time.                                               | Minor |   | X |   |  |
| n/a | 8510 | Always use dbms_application_info to track program process transiently                                                                  | Minor |   | X |   |  |

1. We see a table and a view as a collection. A jar containing beans is labeled "beans". In Java we call such a collection also "beans" (`List<Bean> beans`) and name an entry "bean" (`for (Bean bean : beans) {...}`). An entry of a table is a row (singular) and a table can contain an unbounded number of rows (plural). This and the fact that the Oracle database uses the same concept for their tables and views lead to the decision to use the plural to name a table or a view.
2. Tabs are not used because the indentation depends on the editor configuration. We want to ensure that the code looks the same, independent of the editor used. Hence, no tabs. But why not use 8 spaces? That's the traditional value for a tab. When writing a package function the code in the body has an indentation of 3. That's 24 characters as a starting point for the code. We think it's too much. Especially if we try to keep a line below 100 or 80 characters. Other good options would be 2 or 4 spaces. We settled for 3 spaces as a compromise. The indentation is still good visible, but does not use too much space.